

Q1 In mathematical logic, which of the following are statements?

- (i) There will be snow in January.
- (ii) What is the time now ?
- (iii) Today is Sunday.
- (iv) You must study Discrete Mathematics.

Choose the correct answer from the code given below :

- (A) (i) and (iii) (B) (i) and (ii)
- (C) (ii) and (iv) (D) (iii) and (iv)

Q2 Match List I with List II and choose the correct answer from the code given below.

List I	List II
(a) Equivalence	(i) $p \Rightarrow q$
(b) Contrapositive	(ii) $p \Rightarrow q : q \Rightarrow p$
(c) Converse	(iii) $p \Rightarrow q : \sim p \Rightarrow \sim q$
(d) Implication	(iv) $p \Leftrightarrow q$

- (A) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)
- (B) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)
- (C) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)
- (D) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

Q3 A box contains six red balls and four green balls. Four balls are selected at random from the box. What is the probability that two of the selected balls will be red and two will be green?

- (A) $\frac{1}{14}$ (B) $\frac{3}{7}$
- (C) $\frac{1}{35}$ (D) $\frac{1}{9}$

Q4 A survey has been conducted on methods of commuter travel. Each respondent was asked to check Bus, Train or Automobile as a major method of travelling to work. More than one answer was permitted. The results reported were as follows :

Bus 30 people; Train 35 people; Automobile 100 people; Bus and Train 15 people; Bus and Automobile 15 people, Train and Automobile 20 people; and all the three methods 5 people. How many people completed the survey form?

- (A) 120 (B) 165

- (C) 160 (D) 115

Q5 Which of the following statements are true?

- (i) Every logic network is equivalent to one using just NAND gates or just NOR gates.
 - (ii) Boolean expressions and logic networks correspond to labelled acyclic digraphs.
 - (iii) No two Boolean algebras with n atoms are isomorphic.
 - (iv) Non-zero elements of finite Boolean algebras are not uniquely expressible as joins of atoms.
- Choose the correct answer from the code given below :

- (A) (i) and (iv) only
- (B) (i), (ii) and (iii) only
- (C) (i) and (ii) only
- (D) (ii), (iii) and (iv) only

Q6 The relation $\leq$ and $<$ on a boolean algebra are defined as :

- $x \leq y$ if and only if $x \vee y = y$
- $x < y$ means $x \leq y$ but $x \neq y$
- $x \geq y$ means $y \leq x$ and
- $x > y$ means $y < x$

Considering the above definitions, which of the following is not true in the boolean algebra?

- (i) If $x \leq y$ and $y \leq z$, then $x \leq z$
- (ii) If $x \leq y$ and $y \leq x$, then $x = y$
- (iii) If $x < y$ and $y < z$, then $x \leq y$
- (iv) If $x < y$ and $y < z$, then $x < y$

Choose the correct answer from the code given below :

- (A) (iii) only
- (B) (iv) only
- (C) (i) and (ii) only
- (D) (ii) and (iii) only

Q7 The boolean expression $\overline{A} \cdot B + A \cdot \overline{B} + A \cdot B$ is equivalent to

- (A) $\overline{A} \cdot B$
- (B) $\overline{A} + B$
- (C) $A \cdot B$
- (D) $A + B$



Q8 In PERT/CPM, the merge event represents ___ of two or more events.

- (A) completion (B) beginning
(C) splitting (D) joining

Q9 Use Dual Simplex Method to solve the following problem :

$$\text{Maximize } z = -2x_1 - 3x_2$$

subject to:

$$\begin{aligned} x_1 + x_2 &\geq 2 \\ 2x_1 + x_2 &\leq 10 \\ x_1 + x_2 &\leq 8 \\ x_1, x_2 &\geq 0 \end{aligned}$$

- (A) $x_1 = 2, x_2 = 0$, and $z = -4$
(B) $x_1 = 2, x_2 = 6$ and $z = -22$
(C) $x_1 = 0, x_2 = 2$ and $z = -6$
(D) $x_1 = 6, x_2 = 2$ and $z = -18$

Q10 In computers, subtraction is generally carried out by

- (A) 9's complement
(B) 1's complement
(C) 10's complement
(D) 2's complement

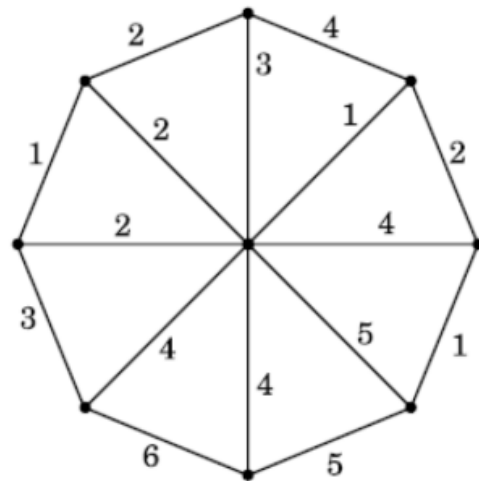
Q11 Consider the following boolean equations :

- (i) $wx + w(x + y) + x(x + y) = x + wy$
(ii) $(w\bar{x}(y + x\bar{z}) + \bar{w}\bar{x})y = \bar{x}y$

What can you say about the above equations?

- (A) (i) is true and (ii) is false
(B) (i) is false and (ii) is true
(C) Both (i) and (ii) are true
(D) Both (i) and (ii) are false

Q12 Consider the graph shown below :



Use Kruskal's algorithm to find the minimum spanning tree of the graph. The weight of this minimum spanning tree is

- (A) 17 (B) 14
(C) 16 (D) 13

Q13 Consider the following statements :

- (i) Auto increment addressing mode is useful in creating self-relocating code.
(ii) If auto increment addressing mode is included in an instruction set architecture, then an additional ALU is required for effective address calculation.
(iii) In auto increment addressing mode, the amount of increment depends on the size of the data item accessed.

Which of the above statements is/are true?

Choose the correct answer from the code given below :

- (A) (i) and (ii) only
(B) (ii) and (iii) only
(C) (iii) only
(D) (ii) only

Q14 A computer uses a memory unit with 256 K words of 32 bits each. A binary instruction code is stored in one word of memory. The instruction has four parts : an indirect bit, an operation code and a register code part to specify one of 64 registers and an address part. How many bits are there in the operation code, the register code part and the address part ?

- (A) 7,6,18 (B) 6,7,18
(C) 7,7,18 (D) 18,7,7



Q15 Consider the following x86 - assembly language instructions :

MOV AL, 153

NEG AL

The contents of the destination register AL (in 8-bit binary notation), the status of Carry Flag (CF) and Sign Flag (SF) after the execution of above instructions, are

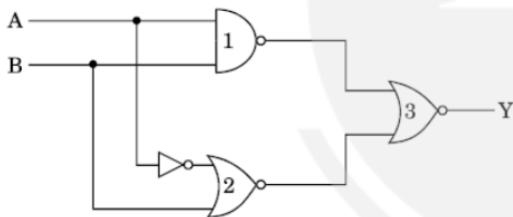
- (A) AL = 01100110; CF = 0; SF = 0
- (B) AL = 01100111; CF = 0; SF = 1
- (C) AL = 01100110; CF = 1; SF = 1
- (D) AL = 01100111; CF = 1; SF = 0

Q16 The decimal floating point number -40.1 represented using IEEE-754 32-bit representation and written in hexadecimal form is

- (A) 0 x C 2206666
- (B) 0 x C 2206000
- (C) 0 x C 2006666
- (D) 0 x C 2006000

Q17 Find the boolean expression for the logic circuit shown below :

(1-NAND gate, 2-NOR gate, 3-NOR gate)



- (A) $A\bar{B}$
- (B) $\bar{A}\bar{B}$
- (C) AB
- (D) $\bar{A}B$

Q18 Consider a disk pack with 32 surfaces, 64 tracks and 512 sectors per pack. 256 bytes of data are stored in a bit serial manner in a sector. The number of bits required to specify a particular sector in the disk is

- (A) 18
- (B) 19
- (C) 20
- (D) 22

Q19 Consider a system with 2 level cache. Access times of Level 1 cache, Level 2 cache and main memory are 0.5 ns, 5 ns and 100 ns respectively. The hit rates of Level 1 and Level 2 caches are 0.7 and 0.8, respectively. What is the average access time of the system ignoring the

search time within the cache?

ptions :

- (A) $35 \cdot 20$ ns
- (B) $7 \cdot 55$ ns
- (C) $20 \cdot 75$ ns
- (D) $24 \cdot 35$ ns

Q20 If a graph (G) has no loops or parallel edges, and if the number of vertices (n) in the graph is $n \geq 3$, then graph G is Hamiltonian if

- (i) $\deg(v) \geq \frac{n}{3}$ for each vertex v
- (ii) $\deg(v) + \deg(w) \geq n$ whenever v and w are not connected by an edge.
- (iii) $E(G) \geq \frac{1}{3}(n-1)(n-2) + 2$

Choose the correct answer from the code given below :

- (A) (i) and (iii) only
- (B) (ii) only
- (C) (ii) and (iii) only
- (D) (iii) only

Q21 The solution of recurrence relation :

$$T(n) = 2T(\sqrt{n}) + \lg(n) \text{ is}$$

- (A) $O(\lg(n))$
- (B) $O(n \lg(n))$
- (C) $O(\lg(n) \lg(n))$
- (D) $O(\lg(n) \lg(\lg(n)))$

Q22 The elements 42,25,30,40,22,35,26 are inserted one by one in the given order into a max-heap.

The resultant max-heap is stored in an array implementation as

- (A) <42, 40, 35, 25, 22, 26, 30>
- (B) <42, 35, 40, 22, 25, 30, 26>
- (C) <42, 40, 35, 25, 22,30,26 >
- (D) <42, 35, 40, 22, 25, 26, 30>

Q23 Consider two sequences X and Y :

$$X = \langle 0, 1, 2, 1, 3, 0, 1 \rangle$$

$$Y = \langle 1, 3, 2, 0, 1, 0 \rangle$$

The length of longest common subsequence between X and Y is

- (A) 2
- (B) 3
- (C) 4
- (D) 5

Q24 Consider the following postfix expression with single-digit operands :



$$6 \ 2 \ 3 \ * / \ 4 \ 2 \ * + 68 \ * -$$

The top two elements of the stack after the second * is evaluated, are:

- (A) 8,2 (B) 8,1
(C) 6,2 (D) 6,3

Q25 A binary search tree is constructed by inserting the following numbers in order :

60,25,72,15,30,68,101,13,18,47,70,34

The number of nodes in the left subtree is

- (A) 5 (B) 6
(C) 7 (D) 3

Q26 In a ternary tree, the number of internal nodes of degree 1, 2, and 3 is 4, 3, and 3 respectively. The number of leaf nodes in the ternary tree is

- (A) 9 (B) 10
(C) 11 (D) 12

Q27 Match List I with List II and choose the correct answer from the code given below.

List I	List II
(Graph Algorithm)	(Time Complexity)
(a) Dijkstra's algorithm	$\Theta(E \log E)$
(b) Kruskal's algorithm	$\Theta(V^3)$
(c) Floyd-Warshall algorithm	$\Theta(V^2)$
(d) Topological sorting	$\Theta(V+E)$

Where V and E are the number of vertices and edges in the graph respectively.

- (A) (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)
(B) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)
(C) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)
(D) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

Q28 In K-coloring of an undirected graph

$G = (V, E)$ is a function

$c : V \rightarrow \{0, 1, \dots, K-1\}$ such that

$c(u) \neq c(v)$ for every edge $(u, v) \in E$.

Which of the following is not correct?

- (A) G is bipartite
(B) G is 2-colorable
(C) G has cycles of odd length
(D) G has no cycles of odd length

Q29 Consider a singly linked list. What is the worst case time complexity of the best-known

algorithm to delete the node a, pointer to this node is q, from the list?

- (A) $O(n \log n)$ (B) $O(n)$
(C) $O(\log n)$ (D) $O(1)$

Q30 The second smallest of n elements can be found with _____ comparisons in the worst case.

- (A) $n - 1$
(B) $(\log n)$
(C) $n + \text{ceil}(\log n) - 2$
(D) $3n/1$

Q31 Let $r = a(a+b)^*$, $s = aa^*b$ and $t = a*b$ be three regular expressions. Consider the following :

- (i) $L(s) \subseteq L(r)$ and $L(s) \subseteq L(t)$
(ii) $L(r) \subseteq L(s)$ and $L(s) \subseteq L(t)$

Choose the correct answer from the code given below :

- (A) Only (i) is correct.
(B) Only (ii) is correct.
(C) Both (i) and (ii) are correct.
(D) Neither (i) nor (ii) is correct.

Q32 Consider the language L given by

$$L = \{2^{nk} \mid k$$

> 0 , and n is non-negative integer number

The minimum number of states of finite automaton that accepts the language L is

- (A) n (B) $n + 1$
(C) $\frac{n(n+1)}{2}$ (D) 2^n

Q33 The number of substrings that can be formed from a string given by a d e f b g h n m p is

- (A) 10 (B) 45
(C) 55 (D) 56

Q34



Consider the following two languages :

$L_1 = \{x \mid \text{for some } y \text{ with } |y| = 2|x|, xy \in L \text{ and } L \text{ is regular language} \}$

$L_2 = \{x \mid \text{for some } y \text{ such that } |x| = |y|, xy \in L \text{ and } L \text{ is regular language} \}$

Which one of the following is correct?

- (A) Only L_1 is regular language
- (B) Only L_2 is regular language
- (C) Both L_1 and L_2 are regular languages
- (D) Both L_1 and L_2 are not regular languages

Q35 Consider the following languages :

$L_1 = \{a^{n+m}b^n a^m \mid n, m \geq 0\}$

$L_2 = \{a^{n+m}b^{n+m}a^{n+m} \mid n, m \geq 0\}$

Which one of the following is correct?

- (A) Only L_1 is context free language
- (B) Only L_2 is context free language
- (C) Both L_1 and L_2 are context free languages
- (D) Both L_1 and L_2 are not context free languages

Q36 Consider R to be any regular language and L_1, L_2 be any two context-free languages.

Which one of the following is correct?

- (A) $\overline{L_1}$ is context free
- (B) $\overline{(L_1 \cup L_2)} - R$ is context free
- (C) $L_1 \cap L_2$ is context free
- (D) $L_1 - R$ is context free

Q37 Consider the following problems :

- (i) Whether a finite state automaton halts on all inputs ?
- (ii) Whether a given context free language is regular?
- (iii) Whether a Turing machine computes the product of two numbers?

Which one of the following is correct?

- (A) Only (i) and (iii) are undecidable problems
- (B) Only (ii) and (iii) are undecidable problems
- (C) Only (i) and (ii) are undecidable problems
- (D) (i), (ii) and (iii) are undecidable problems

Q38 Which of the following problems is decidable for recursive languages (L)?

(A) Is $L = \phi$?

(B) Is $w \in L$, where w is a string ?

(C) 91394342351. Is $L = \Sigma^*$?

(D) Is $L = R$, where R is a given regular set ?

Q39 Consider the following grammar G :

$S \rightarrow A \mid B$

$A \rightarrow a \mid c$

$B \rightarrow b \mid c$

where $\{S, A, B\}$ is the set of non-terminals, $\{a, b, c\}$ is the set of terminals.

Which of the following statement(s) is/are correct?

S_1 : LR(1) can parse all strings that are generated using grammar G .

S_2 : LL(1) can parse all strings that are generated using grammar G .

Choose the correct answer from the code given below :

- (A) Only S_1
- (B) Only S_2
- (C) Both S_1 and S_2
- (D) Neither S_1 nor S_2

Q40 The grammar $S \rightarrow (S) | SS | \epsilon$ is not suitable for predictive parsing because the grammar is

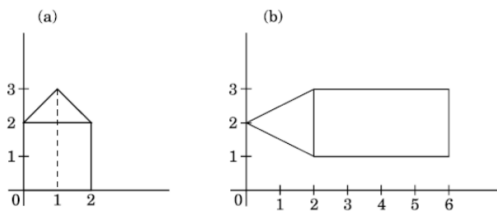
- (A) Right recursive
- (B) Left recursive
- (C) Ambiguous
- (D) An operator grammar

Q41 If the frame buffer has 10-bits per pixel and 8-bits are allocated for each of the R, G, and B components, then what would be the size of the color lookup table (LUT) ?

- (A) $(2^8 + 2^9)$ bytes
- (B) $(2^{10} + 2^8)$ bytes
- (C) $(2^{10} + 2^{24})$ bytes
- (D) $(2^{10} + 2^{11})$ bytes

Q42 Which homogeneous 2D matrix transforms the figure (a) on the left side to the figure (b) on the right ?





- (A) $\begin{pmatrix} 0 & 2 & -6 \\ 2 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}$ (B) $\begin{pmatrix} 0 & -2 & 6 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}$
- (C) $\begin{pmatrix} 1 & -2 & 6 \\ 1 & 0 & 2 \\ 0 & 0 & 1 \end{pmatrix}$ (D) $\begin{pmatrix} 0 & 2 & 6 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}$

Q43 Consider the midpoint (or Bresenham) algorithm for rasterizing lines given below :

- (1) Input (x_1, y_1) and (x_2, y_2)
- (2) $y = y_1$
- (3) $d = f(x_1 + 1, y_1 + 1/2) // f$ is the implicit form of a line
- (4) for $x = x_1$ to x_2
- (5) do
- (6) plot (x, y)
- (7) if $(d < 0)$
- (8) then
- (9) $y = y + 1$
- (10) $d = d + (y_1 - y_2) + (x_2 - x_1)$
- (11) else
- (12) $d = d + (y_1 - y_2)$
- (13) end
- (14) end

Which statements are true?

P : For a line with slope $m > 1$, we should change the outer loop in line (4) to be over y .

Q: Lines (10) and (12) update the decision variable d through an incremental evaluation of the line equation f .

R : The algorithm fails if d is ever 0.

Choose the correct answer from the code given below :

- (A) P only (B) P and Q only
(C) Q and R only (D) P, Q and R

Q44 In 3D Graphics, which of the following statements about perspective and parallel projection is/are true?

P : In a perspective projection, the farthest an object is from the centre of projection, the

smaller it appears.

Q : Parallel projection is equivalent to a perspective projection where the viewer is standing infinitely far away.

R : Perspective projections do not preserve straight lines.

Choose the correct answer from the code given below :

- (A) P and Q only (B) P and R only
(C) Q and R only (D) P, Q and R

Q45 In 3D Graphics, which of the following statements is/are true?

P : Back-face culling is an example of an image-precision visible-surface determination procedure.

Q : Z-buffer is a 16-bit, 32-bit, or 64-bit field associated with each pixel in a frame buffer that can be used to determine the visible surfaces at each pixel.

Choose the correct answer from the code given below :

- (A) P only
(B) Q only
(C) P and Q
(D) Neither P nor Q

Q46 Consider the following pseudo-code fragment, where m is a non-negative integer that has been initialized:

```
p = 0;
k = 0;
while (k < m)
    p = p + 2k;
    k = k + 1;
end while
```

Which of the following is a loop invariant for the while statement?

(Note : a loop invariant for a while statement is an assertion that is true each time the guard is evaluated during the execution of the while statement).

- (A) $p = 2^k - 1$ and $0 \leq k < m$
(B) $p = 2^{k+1} - 1$ and $0 \leq k < m$
(C)



$$p = 2^k - 1 \text{ and } 0 \leq k \leq m$$

$$(D) p = 2^{k+1} - 1 \text{ and } 0 \leq k \leq m$$

- Q47** Consider the following two C++ programs P1 and P2 and two statements S1 and S2 about these programs :

P1	P2
<pre>void f(int a, int *b, int &c) { a = 1; *b = 2; c = 3; } int main() { int i = 0; f(i, &i, i); cout << i; }</pre>	<pre>double a = 1, b = 2; double &f(double &d) { d = 4; return b; } int main() { f(a) = 5; cout << a << " " << b; }</pre>

S1: P1 prints out 3

S2: P2 prints out 4 : 2

What can you say about the statements S1 and S2?

- (A) Only S1 is true.
- (B) Only S2 is true.
- (C) Both S1 and S2 are true.
- (D) Neither S1 nor S2 is true.

- Q48** Consider the following recursive Java function f that takes two long arguments and returns a float value :

```
public static float f(long m, long n)
{
float result =(float )m/(float )n;
if (m<0||n<0)
return 0.0f ;
else
result +=f(m*2,n*3);
return result;
}
```

Which of the following integers best approximates the value of f(2,3)?

- (A) 0
- (B) 1
- (C) 2
- (D) 3

Q49

What does the following Java function perform?

(Assume int occupies four bytes of storage)

```
public static int f(
int a)
{ // Pre-conditions : a>0 and no
overflow/underflow occurs int b=0 ;
for (int i=0; i<32; i++)
{
b=b<<1 ;
b=b | (a & 1);
a=a>>1;// This is a logical shift
}
return b;
}
```

- (A) Returns the int that has the binary representation of integer a.
- (B) Return the int that has the reversed binary representation of integer a.
- (C) Return the int that represents the number of 1's in the binary representation of integer a.
- (D) Return the int that represents the number of 0's in the binary representation of integer a.

- Q50** Which of the following HTML5 codes will affect the horizontal as well as vertical alignment of the table content?

- (A) <td halign="middle" valign="center"> BASH <td >
- (B) <td align="middle" valign="center"> BASH </td>
- (C) <td style="horizontal-align : center; vertical-align : middle;">BASH <td>
- (D) <td style="text-align : center; vertical-align : middle;"> BASH </td>

- Q51** Consider the C / C++ function f() given below :

```
void f(char [w])
{
int x=strlen(w); //length of a string
char c;
for (int i = 0; i<x; i++)
{
c = w[i];
w[i] = w[x-i-1];
w[x-i-1] = c;
}
```



}

Which of the following is the purpose of f()?

- (A) It outputs the contents of the array in reverse order.
 (B) It outputs the contents of the array in the original order.
 (C) It outputs the contents of the array with the characters shifted over by one position.
 (D) It outputs the contents of the array with the characters rearranged so they are no longer recognized as the words in the original phrase.

Q52 The Software Requirement Specification (SRS) is said to be _____ if and only if no subset of individual requirements described in it conflict with each other

- (A) Correct (B) Consistent
 (C) Unambiguous (D) Verifiable

Q53 Software products need perfective maintenance for which of the following reasons?

- (A) To rectify bugs observed while the system is in use.
 (B) When the customers need the product to run on new platforms.
 (C) To support the new features that users want it to support.
 (D) To overcome wear and tear caused by the repeated use of the software.

Q54 Match each UML diagram in List I to its appropriate description in List II.

	List I		List II
(a)	State Diagram	(i)	Describes how the external entities (people, devices) can interact with the system.
(b)	Use-Case Diagram	(ii)	Used to describe the static or structural view of a system.
(c)	Class Diagram	(iii)	Used to show the flow of a business process, the steps of a use-case or the logic of an object's behaviour.
(d)	Activity Diagram	(iv)	Used to describe the dynamic behaviour of

objects and could also be used to describe the entire system's behaviour.

- (A) (a)-(i), (b)-(iv), (c)-(ii), (d)-(iii)
 (B) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)
 (C) (a)-(i), (b)-(iv), (c)-(iii), (d)-(ii)
 (D) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)

Q55 Which of the following statements is/are false?

P: The clean-room strategy for software engineering is based on the incremental software process model.

Q: The clean-room strategy to software engineering is one of the ways to overcome "unconscious" copying of copyrighted code.

Choose the correct answer from the code given below :

- (A) P only
 (B) Q only
 (C) Both P and Q
 (D) Neither P nor Q

Q56 Consider the following method :

```
int f (int m, int n, boolean x, boolean y )
{
    int res = 0;
    if (m < 0){ res = n - m;
    else if (x||y){
        res = -1;
        if (n == m){ res = 1;
    }
    else { res = n; }
    return res;
}/* end of f*
...
```

If P is the minimum number of tests to achieve full statement coverage for f(), and Q is the minimum number of tests to achieve full branch coverage for f(), then (P, Q) =

- (A) (3,4) (B) (4,3)
 (C) (2,3) (D) (3,2)

Q57 A legacy software system has 940 modules. The latest release required that 90 of these modules be changed. In addition, 40 new modules were



added and 12 old modules were removed.
Compute the software maturity index for the system.

- (A) 0.849 (B) 0.524
(C) 0.725 (D) 0.923

Q58 Which of the following statements is/are true?
P : Software Reengineering is preferable for software products having high failure rates, poor design, and/or poor code structure.

Q: Software Reverse Engineering is the process of analyzing software to recover its design and requirements specification.

Choose the correct answer from the code given below :

- (A) P only
(B) Q only
(C) Both P and Q
(D) Neither P nor Q

Q59 Which of the following is not one of the principles of agile software development method?

- (A) Customer involvement
(B) Embrace change
(C) Incremental delivery
(D) Following the plan

Q60 Software coupling involves dependencies among pieces of software called modules. Which of the following are correct statements concerning module coupling?

P: Common coupling occurs when two modules share the same global data.

Q: Control coupling occurs when modules share a composite data structure and use only parts of it.

R: Content coupling occurs when one module modifies or relies on the internal working of another module.

Choose the correct answer from the code given below :

- (A) P and Q only
(B) P and R only
(C) Q and R only
(D) All of P, Q, and R

Q61

Match the following Secret Key Algorithms (List I) with the Corresponding Key Lengths (List II) and choose the correct answer from the code given below.

	List I		List II
(a)	Blowfish	(i)	128 - 256 bits
(b)	DES	(ii)	128 bits
(c)	IDEA	(iii)	1- 448 bits
(d)	RC5	(iv)	56 bits

- (A) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
(B) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)
(C) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)
(D) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

Q62 The four byte IP Address consists of
(A) Network Address
(B) Host Address
(C) Both Network and Host Addresses
(D) Neither Network nor Host Address

Q63 Consider ISO-OSI network architecture reference model. Session layer of this model offers dialog control, token management and _____ as services.
(A) Synchronization
(B) Asynchronization
(C) Flow Control
(D) Errors

Q64 Suppose a cloud contains software stack such as Operating systems, Application softwares, etc. This model is referred as _____ model.
(A) SaaS (B) PaaS
(C) IaaS (D) MaaS

Q65 The Third Generation mobile phones are digital and based on
(A) AMPS (B) D-AMPS
(C) CDMA (D) Broadband CDMA

Q66 Which of the following statement/s is/are true?
(i) Firewalls can screen traffic going into or out of an organization.
(ii) Virtual private networks can simulate an old leased network to provide certain desirable properties.



Choose the correct answer from the code given below :

- (A) (i) only
- (B) (ii) only
- (C) Both (i) and (ii)
- (D) Neither (i) nor (ii)

Q67 Which of the following statement/s is/are true?

- (i) Windows XP supports both peer-peer and client-server networks.
- (ii) Windows XP implements Transport protocols as drivers that can be loaded and unloaded from the system dynamically.

Choose the correct answer from the code given below :

- (A) (i) only
- (B) (ii) only
- (C) Neither (i) nor (ii)
- (D) Both (i) and (ii)

Q68 In Linux operating system environment _____ command is used to print a file.

- (A) print (B) ptr
- (C) pr (D) lpr

Q69 To overcome difficulties in Readers-Writers problem, which of the following statement/s is/are true?

- (i) Writers are given exclusive access to shared objects.
- (ii) Readers are given exclusive access to shared objects.
- (iii) Both Readers and Writers are given exclusive access to shared objects.

Choose the correct answer from the code given below :

- (A) (i) only
- (B) (ii) only
- (C) (iii) only
- (D) Both (ii) and (iii)

Q70 Suppose a system has 12 instances of some resource with n processes competing for that resource. Each process may require 4 instances of the resource. The maximum value of n for which the system never enters into deadlock is

- (A) 3 (B) 4
- (C) 5 (D) 6

Q71 Dirty bit is used to show the

- (A) wrong page
- (B) page with corrupted data
- (C) page with low frequency occurrence
- (D) page that is modified after being loaded into cache memory

Q72 Suppose for a process P , reference to pages in order are (1,2,4,5,2,1,2,4). Assume that main memory can accommodate 3 pages and the main memory has already pages 1 and 2 in the order 1 - first, 2 - second. At this moment, assume FIFO Page Replacement Algorithm is used then the number of page faults that occur to complete the execution of process P is

- (A) 4 (B) 3
- (C) 5 (D) 6

Q73 Consider the following set of processes and the length of CPU burst time given in milliseconds :

Process	CPU Burst Time(ms)
P_1	5
P_2	7
P_3	6
P_4	4

Assume that processes being scheduled with Round-Robin Scheduling Algorithm with time quantum 4ms. Then the waiting time for P_4 is _____ ms.

- (A) 0 (B) 4
- (C) 12 (D) 6

Q74 Suppose P , Q , and R are cooperating processes satisfying Mutual Exclusion conditions. Then, if the process Q is executing in its critical section then

- (A) Both ' P ' and ' R ' execute in the critical section.
- (B) Neither ' P ' nor ' R ' executes in their critical section.
- (C) ' P ' executes in the critical section.
- (D) ' R ' executes in the critical section.

Q75 A process residing in Main Memory and Ready and Waiting for execution is kept on?

- (A) Execution Queue
- (B) Job Queue



- (C) Ready Queue
(D) Wait Queue

Q76 _____ system call creates new process in Unix.

- (A) create (B) create new
(C) fork (D) fork new

Q77 An attribute A of datatype varchar(20) has the value 'xyz', and the attribute B of datatype char(20) has the value "lmnop", then the attribute A has _____ spaces and attribute B has _____ spaces.

- (A) 3,5 (B) 20,20
(C) 3,20 (D) 20,5

Q78 Data warehouse contains _____ data that is never found in the operational environment.

- (A) Summary (B) Encoded
(C) Encrypted (D) Scripted

Q79 _____ command is used to remove a relation from an SQL database.

- (A) Drop table (B) Delete table
(C) Remove table (D) Update table

Q80 Which of the following statement/s is/are true?

- (i) Facebook has the world's largest Hadoop Cluster.
(ii) Hadoop 2.0 allows live stream processing of Real time data.

Choose the correct answer from the code given below :

- (A) (i) only
(B) (ii) only
(C) Both (i) and (ii)
(D) Neither (i) nor (ii)

Q81 Data scrubbing is

- (A) A process to reject data from the data warehouse and to create the necessary indexes.
(B) A process to load the data in the data warehouse and to create the necessary indexes.
(C) A process to upgrade the quality of data after it is moved into a data warehouse.
(D)

A process to upgrade the quality of data before it is moved into a data warehouse.

Q82 Consider a relation schema

$R = (A, B, C, D, E, F)$ on which the following functional dependencies hold :

- $A \rightarrow B$
 $B, C \rightarrow D$
 $E \rightarrow C$
 $D \rightarrow A$

What are the candidate keys of R?

- (A) AE and BE
(B) AE, BE, and DE
(C) AEF, BEF and BCF
(D) AEF, BEF and DEF

Q83 Consider the following sequence of two transactions on a bank account (A) with an initial balance of 20,000 that transfers 5,000 to another account (B) and then applies 10% interest.

- (i) T1 start
(ii) T1 A old = 20,000 new 15,000
(iii) T1 B old = 12,000 new = 17,000
(iv) T1 commit
(v) T2 start
(vi) T2 A old = 15,000 new = 16,500
(vii) T2 commit

Suppose the database system crashes just before log record (vii) is written. When the system is restarted, which one statement is true of the recovery process?

- (A) We must redo log record (vi) to set A to 16,500 .
(B) We must undo log record (vi) to set A to 16,500 and then redo log records (ii) and (iii).
(C) We need not redo log records (ii) and (iii) because transaction T1 has committed.
(D) We can apply redo and undo operations in arbitrary order because they are idempotent.

Q84 Consider the following tables (relations) :

Students	
Roll-No	Name
18CS101	Ramesh
18CS102	Mukesh



ISCS103	Ramesh
---------	--------

Performance		
Roll-No	Course	Marks
18CS101	DBMS	60
18CS101	Compiler Design	65
18CS102	DBMS	80
18CS103	DBMS	85
18CS102	Compiler Design	75
18CS103	Operating System	70

Primary keys in the tables are shown using underline. Now, consider the following query :

```
SELECT S.Name, Sum (P.Marks)
FROM Students S, Performance P
WHERE S.Roll-No = P.Roll-No
GROUP BY S.Name
```

The number of rows returned by the above query is

- (A) 0 (B) 1
(C) 2 (D) 3

Q85 A clustering index is defined on the fields which are of type

- (A) non-key and ordering
(B) non-key and non-ordering
(C) key and ordering
(D) key and non-ordering

Q86 A host is connected to a department network which is part of a university network. The university network, in turn, is part of the Internet. The largest network, in which the Ethernet address of the host is unique, is

- (A) the Internet
(B) the university network
(C) the department network
(D) the subnet to which the host belongs

Q87 An Internet Service Provider (ISP) has following chunk of CIDR-based IP addresses available with it : 245.248.128.0/20. The ISP wants to give half of this chunk of addresses to organization A and

a quarter to organization B while retaining the remaining with itself. Which of the following is a valid allocation of addresses to A and B ?

- (A) 245.248.136.0/21 and 245.248.128.0/22
(B) 245.248.128.0/21 and 245.248.128.0/22
(C) 245.248.132.0/22 and 245.248.132.0/21
(D) 245.248.136.0/24 and 245.248.132.0/21

Q88 Consider the following two statements :

S1 : TCP handles both congestion and flow control

S2 : UDP handles congestion but not flow control

Which of the following options is correct with respect to the above statements (S1) and (S2)?

- (A) Neither S1 nor S2 is correct
(B) S1 is not correct but S2 is correct
(C) S1 is correct but S2 is not correct
(D) Both, S1 and S2 are correct

Q89 Identify the correct sequence in which the following packets are transmitted on the network by a host when a browser requests a webpage from a remote server, assuming that the host has just been restarted.

- (A) HTTP GET request, DNS query, TCP SYN
(B) DNS query, HTTP GET request, TCP SYN
(C) TCP SYN, DNS query, HTTP GET request
(D) DNS query, TCP SYN, HTTP GET request

Q90 Suppose that everyone in a group of N people wants to communicate secretly with the N-1 others using symmetric key cryptographic system. The communication between any two persons should not be decodable by the others in the group. The number of keys required in the system as a whole to satisfy the confidentiality requirement is

- (A) $N(N-1)$ (B) $N(N-1)/2$
(C) $2N$ (D) $(N-1)^2$

Q91 An agent can improve its performance by

- (A) Perceiving (B) Responding
(C) Learning (D) Observing

Q92 Which of the following is true for semi-dynamic environment?

- (A) The environment may change while the agent is deliberating.



(C) 5

(D) 6

- Q98** Consider a vocabulary with only four propositions A, B, C, and D. How many models are there for the following sentence?

$$\neg A \vee \neg B \vee \neg C \vee \neg D$$

(A) 7

(B) 8

(C) 15

(D) 16

- Q99** Consider the sentence below.

"There is a country that borders both India and Nepal."

Which of the following represents the above sentence correctly?

(A) $\exists \text{Country}(c) \wedge \text{Border}(c, \text{India}) \wedge \text{Border}(c, \text{Nepal})$

(B) $\exists \text{Country}(c) \Rightarrow [\text{Border}(c, \text{India}) \wedge \text{Border}(c, \text{Nepal})]$

(C) $[\exists \text{Country}(c)] \Rightarrow [\text{Border}(c, \text{India}) \wedge \text{Border}(c, \text{Nepal})]$

(D) $\exists \text{Border}(\text{Country}(c), \text{India} \wedge \text{Nepal})$

- Q100** A full joint distribution for the Toothache, Cavity, and Catch is given in the table below:

	Toothache	Toothache	$\neg$ Toothache	$\neg$ Toothache
	Catch	$\neg$ Catch	Catch	$\neg$ Cathe
Cavity	0.108	0.012	0.072	0.008
$\neg$ Cavity	0.016	0.064	0.144	0.576

What is the probability of Cavity, given evidence of Toothache?

(A) $\langle 0.2, 0.8 \rangle$

(B) $\langle 0.4, 0.8 \rangle$

(C) $\langle 0.6, 0.8 \rangle$

(D) $\langle 0.6, 0.4 \rangle$



Answer Key

Q1 (A)
Q2 (D)
Q3 (B)
Q4 (A)
Q5 (C)
Q6 (A)
Q7 (D)
Q8 (A)
Q9 (A)
Q10 (D)
Q11 (C)
Q12 (C)
Q13 (C)
Q14 (A)
Q15 (A)
Q16 (A)
Q17 (C)
Q18 (C)
Q19 (B)
Q20 (B)
Q21 (D)
Q22 (C)
Q23 (C)
Q24 (B)
Q25 (C)
Q26 (B)
Q27 (B)
Q28 (C)
Q29 (B)
Q30 (C)
Q31 (A)
Q32 (B)

Q33 (D)
Q34 (C)
Q35 (A)
Q36 (D)
Q37 (B)
Q38 (B)
Q39 (D)
Q40 (C)
Q41 (D)
Q42 (B)
Q43 (B)
Q44 (A)
Q45 (B)
Q46 (B)
Q47 (A)
Q48 (C)
Q49 (B)
Q50 (D)
Q51 (B)
Q52 (B)
Q53 (C)
Q54 (D)
Q55 (D)
Q56 (A)
Q57 (A)
Q58 (C)
Q59 (D)
Q60 (B)
Q61 (C)
Q62 (C)
Q63 (A)
Q64 (A)



Q65 (D)
Q66 (C)
Q67 (D)
Q68 (D)
Q69 (A)
Q70 (A)
Q71 (D)
Q72 (C)
Q73 (C)
Q74 (B)
Q75 (C)
Q76 (C)
Q77 (C)
Q78 (A)
Q79 (A)
Q80 (C)
Q81 (D)
Q82 (D)

Q83 (B)
Q84 (C)
Q85 (A)
Q86 (A)
Q87 (A)
Q88 (C)
Q89 (D)
Q90 (B)
Q91 (C)
Q92 (B)
Q93 (B)
Q94 (D)
Q95 (D)
Q96 (C)
Q97 (C)
Q98 (C)
Q99 (A)
Q100 (D)



Hints & Solutions

Q1 Text Solution:

Option A is Correct

(i) There will be snow in January.

This is a declarative sentence. Although its truth value depends on the context (e.g., location and weather conditions), it can be classified as true or false in a specific context. Hence, This is a statement.

(ii) What is the time now?

This is a question and does not assert anything that can be true or false. Hence, This is not a statement.

(iii) Today is Sunday.

This is a declarative sentence that asserts something about the day. Depending on the day, it can be classified as true or false. Hence, This is a statement.

(iv) You must study Discrete Mathematics.

This is a command or an advice, not a declarative sentence. It cannot be classified as true or false. Hence, This is not a statement.

Q2 Text Solution:

Option D is Correct

List I	List II	Explanation
(a) Equivalence	(iv) $p \Leftrightarrow q$	Logical equivalence means p and q have the same truth value, represented by $p \Leftrightarrow q$.
(b) Contrapositive	(iii) $\sim q \Rightarrow \sim p$	The contrapositive of $p \Rightarrow q$ is $\sim q \Rightarrow \sim p$. Both have the same truth value.
(c) Converse	(ii) $q \Rightarrow p$	The converse of $p \Rightarrow q$ reverses the implication, becoming $q \Rightarrow p$.
(d) Implication	(i) $p \Rightarrow q$	Implication states that if p is true, then q must also be true, represented by $p \Rightarrow q$.

Q3 Text Solution:

Option B is Correct

4 balls can be selected from 10 balls in $^{10}C_4$ ways.

2 red balls can be selected from 6 red balls in 6C_2 ways.

2 green balls can be selected from 4 green balls in 4C_2 ways.

$$\text{hence probability} = \frac{{}^6C_2 \times {}^4C_2}{{}^{10}C_4} = \frac{3}{7}$$

Q4 Text Solution:

Option A is Correct

Let 'A' denotes people travelled by Bus, $n(A) = 30$ people

Let 'B' denotes people travelled by Train, $n(B) = 35$ people

Let 'C' denotes people travelled by Automobile, $n(C) = 100$ people

Let ' $A \cap B$ ' denotes people travelled by bus and Train, $n(A \cap B) = 15$ people

Let ' $A \cap C$ ' denotes people travelled by bus and Automobile, $n(A \cap C) = 15$ people

Let ' $B \cap C$ ' denotes people travelled by Train and Automobile, $n(B \cap C) = 20$ people

Let ' $A \cap B \cap C$ ' denotes people travelled by Bus, Train and Automobile, $n(A \cap B \cap C) = 5$ people

Number of people completed the survey

$$\begin{aligned} n(A \cup B \cup C) &= n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C) \\ &= 30 + 35 + 100 - 15 - 15 - 20 + 5 \\ &= 120 \end{aligned}$$

Q5 Text Solution:

Explanation:

(i) Every logic network is equivalent to one using just NAND gates or just NOR gates.

TRUE

Reason:

- NAND and NOR are **universal gates**.
- Any Boolean function can be implemented using:
 - Only NAND gates
 - OR only NOR gates

So this statement is **correct**.



(ii) Boolean expressions and logic networks correspond to labelled acyclic digraphs.

TRUE

Reason:

- A logic circuit has:
 - Inputs $\rightarrow$ gates $\rightarrow$ output
- No cycles (feedback not allowed in combinational logic)
- So it forms a **Directed Acyclic Graph (DAG)**

Hence correct.

(iii) No two Boolean algebras with n atoms are isomorphic.

This says:

"No two Boolean algebras with n atoms are isomorphic."

This is **FALSE**

Fact:

- Any two Boolean algebras with the same number of atoms are **isomorphic**.
- A finite Boolean algebra with n atoms has size 2^n .
- All such algebras are structurally identical.

So statement (iii) is wrong.

(iv) Non-zero elements of finite Boolean algebras are not uniquely expressible as joins of atoms.

This says elements are **not uniquely expressible**.

This is FALSE

Fact:

- In a finite Boolean algebra, every element can be uniquely written as a **join (OR) of atoms**.

So statement (iv) is wrong.

Option C – (i) and (ii) only

Q6 Text Solution:

Option A is Correct

(i) If $x \leq y$ and $y \leq z$, then $x \leq z$:

This follows the transitive property of $\leq$. If $xv = y$ and $yv = z$, then $xv = z$, so $x \leq z$. Hence, this statement is true.

(ii) If $x \leq y$ and $y \leq x$, then $x = y$:

If $xv = y$ and $yv = x$, then $x = y$, since the equality condition holds in both directions. Hence, this statement is true.

(iii) If $x < y$ and $y < z$, then $x < z$:

$x < y$ means $x \leq y$ and $x \neq y$, but the second condition ($x \neq y$) is not explicitly stated here. This makes the statement incomplete or misleading. Hence, this statement is false.

(iv) If $x < y$ and $y < z$, then $x < z$:

This is true by the transitive property of $<$. Since $x < y$ (meaning $x \leq y$ and $x \neq y$) and $y < z$ (meaning $y \leq z$ and $y \neq z$), it follows that $x < z$ (i.e., $x \leq z$ and $x \neq z$).

Hence, this statement is true.

The Final Answer is: III Only.

Q7 Text Solution:

Option D is Correct

$$\overline{A}B + A\overline{B} + AB = \overline{A}B + A(\overline{B} + B)$$

$$(\text{since } B + \overline{B} = 1)$$

$$\overline{A}B + A = (\overline{A} + A)(A + B) \quad (\text{Since Distributive Law})$$

$$= A + B$$

Q8 Text Solution:

Option A is Correct

In PERT (Program Evaluation and Review Technique) and CPM (Critical Path Method), a merge event occurs when multiple preceding activities converge at a single node. This implies that all the activities leading to this node must be completed for the merge event to occur.

Thus, in this context:

- A merge event signals the completion of all converging activities, enabling the next activity or set of activities to start.
- The merge event itself does not represent the beginning of any specific activity; it simply ensures that all incoming activities have been finished.

A) Completion (Correct):

A merge event represents the completion of all incoming activities, as all prerequisites must be met before moving forward.

Q9 Text Solution:

Option A is Correct

We convert the constraints into equality form using slack or surplus variables:

$$1. x_1 + x_2 \geq 2 \text{ becomes: } x_1 + x_2 - s_1 = 2, s_1 \geq 0.$$

$$2. 2x_1 + x_2 \leq 10 \text{ becomes: } 2x_1 + x_2 + s_2 = 10, s_2 \geq 0.$$

$$3. x_1 + x_2 \leq 8 \text{ becomes: } x_1 + x_2 + s_3 = 8, s_3 \geq 0.$$



Objective function in standard form:

$$z = 2x_1 + 3x_2 = 0$$

The initial tableau is:

Basis	x_1	x_2	s_1	s_2	s_3	RHS
s_1	1	1	-1	0	0	2
s_2	2	1	0	1	0	10
s_3	1	1	0	0	1	8
Z	-2	-3	0	0	0	0

The Dual Simplex Method focuses on ensuring the RHS (Right-Hand Side) values are non-negative while optimizing the objective function.

Iteration 1

1. Select the leaving variable (based on the most negative RHS): RHS values are: 2, 10, 8. All are non-negative, so the solution is already feasible.
2. Select the entering variable (based on the most negative coefficient in the z-row):
 - The most negative coefficient is -3 (column x_2).
3. Perform the ratio test to find the pivot row:

Ratio = RHS / Pivot Column Coefficient

So, For s_1 : $2/1=2$, For s_2 : $10/1=10$, For s_3 : $8/1=8$.

The smallest ratio is 2 (row s_1).

1. Pivot on (s_1, x_2) = 1.

Iteration 2: Update the Tableau

After pivoting,

Basis	x_1	x_2	s_1	s_2	s_3	RHS
x_2	1	1	-1	0	0	2
s_2	2	0	1	1	0	8
s_3	1	0	1	0	1	6
Z	-2	0	-3	0	0	-6

Iteration 3: Optimality Check

1. Select the entering variable (based on the most negative coefficient in the z-row): The most negative coefficient is -2 (column x_1).
 2. Perform the ratio test to find the pivot row:
- For x_2 : $2/1 = 2$, For s_2 : $8/2 = 4$, and For s_3 : $6/1 = 6$.
The smallest ratio is 2 (row x_2).
1. Pivot on (x_2, x_1) = 1.

Iteration 4: Final Tableau

After pivoting again,

Basis	x_1	x_2	s_1	s_2	s_3	RHS
x_1	1	0	-1	0	0	2
x_2	0	1	0	1	0	0
s_3	0	0	1	0	1	6
Z	0	0	-2	0	0	-4

Hence, Optimal solution: $x_1=2$, and $x_2=0$ and

Maximum value of z:

$$Z = -2(2) - 3(0) = -4$$

Q10 Text Solution:

Option D is Correct

Computers use 2's complement for subtraction due to its simplicity and efficiency in binary arithmetic.

Q11 Text Solution:

Option C is Correct

(i) $wx + w(x+y) + x(x+y) = x + wy$

Taking LHS to solve, $wx + w(x+y) + x(x+y)$:

Since, $w(x+y) = wx + wy$

$(x+y) = x$ (since $x \cdot x = x$ and $x \cdot y = x$).

Substituting these:

$$wx + (wx + wy) + x = wx + wx + wy + x = wx + wy + x = x + wy$$

Hence, (i) is true.

(ii) $(wx'(y+xz') + w'x')y = x'y$

Taking LHS,

$$(wx'y + wx'xz' + w'x')y = wx'yy + wx'xz'y + w'x'y$$

$$= wx'y + wx'xz'y + w'x'y \text{ (Since, } yy=y, wx'yy=wx'y.)$$

$$= x'y (w + w') + 0 \text{ (Since, } w + w' = 1 \text{ and } wx'xz'y = 0 \text{ as } x' \cdot x = 0)$$

$$= x'y$$

Hence, (ii) is also True

Q12 Text Solution:

Option C is Correct

Arranging the weights in ascending order

1, 1, 1, 2, 2, 2, 2, 3, 3, 4, 4, 4, 4, 5, 5, 6

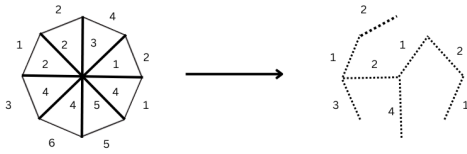
Now, Arranging the weighted edges in such a manner there is no cyclic occurrence and finding the sum for MST, as per the figure below

Iteratively Add Edges to the MST:

- Start adding edges to the MST from the sorted list.



- Add an edge only if it does not form a cycle in the MST.
- Repeat this until all vertices are connected.



$$1+1+1+2+2+2+3+4 = 16$$

Hence, The weight of the Minimum Spanning Tree is 16.

Q13 Text Solution:

Option C is Correct

Statement (i): Auto increment addressing mode is useful in creating self-relocating code.

Auto increment addressing mode automatically updates the address pointer after accessing a memory location. This is useful in iterating through arrays or data structures, but it does not directly facilitate self-relocating code.

Self-relocating code refers to code that can adjust itself to execute properly regardless of where it is loaded in memory. Auto increment addressing does not inherently assist in achieving this.

Hence, This statement is false.

Statement (ii): If auto increment addressing mode is included in an instruction set architecture, then an additional ALU is required for effective address calculation.

In auto increment addressing mode, the effective address is calculated by adding a fixed or variable increment to the current value of the address pointer. This operation requires an arithmetic addition.

An additional ALU may not always be required; this depends on the hardware design. Many processors reuse the existing ALU for both effective address computation and general arithmetic operations.

However, in some architectures, a separate unit may be provided for this purpose.

Hence, This statement is not universally true and is false for most modern architectures.

Statement (iii): In auto increment addressing mode, the amount of increment depends on the

size of the data item accessed.

In most architectures, the auto increment amount depends on the size of the data being accessed (e.g., a 4-byte increment for a 32-bit integer or a 1-byte increment for a character).

This ensures that the next memory address points to the subsequent data item.

Hence, This statement is true.

Q14 Text Solution:

Option A is Correct

1. Word size: 32bits.
2. Memory size: 256K words= 2^{18} word.
3. Indirect bit: 1bit.
4. Number of registers: $n=64$.

Bits needed to address registers:

Register code bits= $\log_2(n)$, where $n=64$.

Since $n=64$, we compute:

Register Code Bits= $\log_2(64) = 6$ bits.

Bits needed to address memory locations:

Address part bits = $\log_2(\text{memory size}) = \log_2(2^{18}) = 18$ bits.

As the memory size is 2^{18} words, the address part requires 18bits.

Remaining Bits for Operation Code (Opcode):

Total instruction size is 32bits,

Opcode Bits = $32 - (\text{Indirect Bit} + \text{Register Code} + \text{Address Part})$

Opcode Bits = $32 - (1+6+18) = 7$ bits.

Hence, Final Allocation:

Part	Bits Required
Indirect Bit	1
Operation Code	7
Register Code	6
Address Part	18

Q15 Text Solution:

Option D is Correct

To solve this problem, let's follow the execution of the assembly instructions step by step and determine the values of the registers and flags.

Step 1: MOV AL, 153

First, the MOV instruction moves the decimal value 153 into the AL register. So, Convert 153 to 8-bit binary: 153 in binary is 10011001



So, after the MOV instruction: AL = 10011001

Step 2: NEG AL

The NEG instruction computes the complement of the operand.

Invert all bits (one's complement): 10011001 → 01100110

So, Carry Flag (CF): There is no carry so CF = 0

And, Sign Flag (SF): The SF reflects the sign of the result. The MSB of AL after the NEG operation is 0. (SF = 0)

Q16 Text Solution:

Option A is Correct

IEEE 754, 32-bit representation, 1 bit for sign, 8 bits for Exponent and 23 bits for mantissa.

Note that Exponent represented in Excess-127 format.

We have to represent -40.1, So Sign bit = 1

$$40 = 32 + 8 = (101000)_2$$

$$0.1 = (00011\ 0011\ 0011\ 0011\ 0011\ 00)_2$$

represented with 23 bits

40.1 = 101000.00011 0011 0011 0011 0011 00 in binary form.

$$= 1.01000\ 00011\ 0011\ 0011\ 0011\ 0011\ 00 \times 2^5$$

Now it is in 1.M format, Keep only 23 bits in Mantissa

$$= 1.01000\ 00011\ 0011\ 0011\ 0011\ 0 \times 2^5$$

NOW it is in 1.M format, and Mantissa have 23 bits only.

True exponent is 5 (which is in power of 2)

$$\text{So, Excess Exponent value} = 5 + 127 = 132 = 128 + 4 = 10000100$$

Hence, Final representation will be:

sign	Exponent	Mantissa
1	10000100	01000000110011001100110

Finally, it is like

$$(11000010001000000110011001100110)_2 \\ 1100\ 0010\ 0010\ 0000\ 0110\ 0110\ 0110\ 0110 = \\ 0xC2206666$$

Q17 Text Solution:

Option C is Correct

$$Y = ((AB)' + (A'B))'$$

$$Y = ((AB)')' \cdot ((A'B)')' \text{ (Applying De Morgan's Law)}$$

$$Y = AB \cdot (A+B) \text{ (Applying De Morgan's Law)}$$

$$Y = ABA' + ABB \text{ (Since, } A.A' = 0 \text{ and } B.B = B)$$

$$Y = AB$$

Q18 Text Solution:

Option C is Correct

$$\text{Total sectors} = \text{No. of surfaces} \times \text{No. of tracks per surface} \times \text{number of sectors per track} \\ = 32 \times 64 \times 512 = 2^5 \times 2^6 \times 2^9 = 2^{20}.$$

∴ No. of bits required to identify each sector = 20 bits

Q19 Text Solution:

Option B is Correct

$$T_{\text{avg access}} = H_1 \cdot T_1 + (1-H_1) (H_2 \cdot T_2 + (1-H_2) \cdot T_M)$$

$$T_{\text{avg access}} = (0.7 \times 0.5) + (0.3)((0.8 \times 5) + (0.2) \cdot 100)$$

$$T_{\text{avg access}} = (0.35) + (0.3)(4+20) = 0.35 + 7.2 = 7.55 \text{ ns}$$

Q20 Text Solution:

Option B is Correct

In an Hamiltonian Graph (G) with no loops and parallel edges:

- According to Dirac's theorem in a n vertex graph, $\deg(v) \geq \frac{n}{2}$ for each vertex of G. So, statement (i) is false.
- According to Ore's theorem $\deg(v) + \deg(w) \geq n$ for every n and v not connected by an edge is sufficient condition for a graph to be hamiltonian. So, statement (ii) is True.
- If $|E(G)| \geq \frac{1}{2} * [(n-1)(n-2)]$ then graph is connected but it doesn't guaranteed to be Hamiltonian Graph. So, statement (iii) is false.

Q21 Text Solution:

Option D is Correct

Let's analyze the given recurrence relation:

$$T(n) = 2T(\sqrt{n}) + \lg(n)$$

To solve this, we can use a substitution technique:

1. Substitute $n = 2^m$: This transforms the recurrence into: $T(2^m) = 2T(2^{m/2}) + m$
2. Define a new function $S(m) = T(2^m)$: This gives us: $S(m) = 2S(m/2) + m$

Now, we have a more standard recurrence relation: $S(m) = 2S(m/2) + m$



This is a classic divide-and-conquer recurrence, which can be solved using the Master Theorem. Applying the Master Theorem where $a = 2$, $b = 2$, and $f(m) = m$

Calculate $\log_b(a)$: $\log_2(2) = 1$

Compare $f(m)$ with $n^{(\log_b(a))}$: $m = n^1$

Since $f(m) = \Theta(n^{(\log_b(a))})$, we are in Case 2 of the Master Theorem.

Therefore, the solution to $S(m)$ is: $S(m) = \Theta(m \log m)$

Substituting back $n = 2^m$: $T(n) = S(\log n) = \Theta(\log n * \log \log n)$

Hence, the correct answer is: $O(\lg(n) \lg(\lg(n)))$

Q22 Text Solution:

Explanation:

Hint: Max Heap = Parent Node > Child Node

Sequence: 42, 25, 30, 40, 22, 35, 26

Step 1: Insert 42 → Heap: [42] → No rearrangement is needed.

Step 2: Insert 25 → Heap: [42, 25] → 25 is the child of 42, No rearrangement is needed since $42 > 25$.

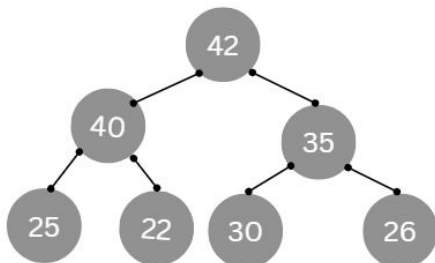
Step 3: Insert 30 → Heap: [42, 25, 30] → 30 is added as a child of 42. → No rearrangement is needed since $42 > 30$.

Step 4: Insert 40 → Heap: [42, 25, 30, 40] → 40 is added as a child of 25. Since $40 > 25$, swap 40 and 25 → Heap: [42, 40, 30, 25].

Step 5: Insert 22 → Heap: [42, 40, 30, 25, 22] → 22 is added as a child of 40. No rearrangement is needed since $40 > 22$.

Step 6: Insert 35 → Heap: [42, 40, 30, 25, 22, 35] → 35 is added as a child of 30. Since $35 > 30$, swap 35 and 30: → Heap: [42, 40, 35, 25, 22, 30].

Step 7: Insert 26 → Heap: [42, 40, 35, 25, 22, 30, 26] → 26 is added as a child of 35 → No rearrangement is needed since $35 > 26$



The Final Answer is: <42, 40, 35, 25, 22, 30, 26 >

Q23 Text Solution:

Option C is Correct

To calculate the length of the longest common subsequence (LCS) between sequences $X = \langle 0, 1, 2, 1, 3, 0, 1 \rangle$ and $Y = \langle 1, 3, 2, 0, 1, 0 \rangle$, we use a dynamic programming approach.

Steps to Solve:

1. Define LCS Table:

Let $L[i][j]$ represent the length of the LCS of the first i elements of X and the first j elements of Y .

2. Initialization:

If $i=0$ or $j=0$, then $L[i][j]=0$ (the LCS of an empty sequence is 0).

3. Recursive Relation:

- If $X[i-1] == Y[j-1]$, then $L[i][j] = L[i-1][j-1] + 1$.
- Otherwise, $L[i][j] = \max(L[i-1][j], L[i][j-1])$.

4. Compute the Table for X and Y :

We will fill the table based on the sequences.

LCS Table for $X = \langle 0, 1, 2, 1, 3, 0, 1 \rangle$ and $Y = \langle 1, 3, 2, 0, 1, 0 \rangle$:

i \ j	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1 (0)	0	0	0	0	1	1	1
2 (1)	0	1	1	1	1	2	2
3 (2)	0	1	1	2	2	2	2
4 (1)	0	1	1	2	2	3	3
5 (3)	0	1	2	2	2	3	3
6 (0)	0	1	2	2	3	3	3
7 (1)	0	1	2	2	3	4	4

The length of the longest common subsequence is $L[7][6] = 4$.

Q24 Text Solution:

Explanation:

- A **stack** is a linear data structure that stores elements in **LIFO (Last In, First Out)** order.
- Insertion and deletion only happen at one end, **called the top of the stack**.

Step 1: Push 6 onto the stack → Stack [6]

Step 2: Push 2 onto the stack → Stack [6, 2]

Step 3: Push 3 onto the stack → Stack [6, 2, 3]

Step 4: Encounter the $*$ operator (Multiply the top elements, 2 and 3) → $2 * 3 = 6$



Push the result (6) back on the stack → Stack [6, 6]

Step 5: Encounter the / operator (divide the top two elements, 6 and 6) → $6/6 = 1$

Push the result(1) back on the stack → Stack [1]

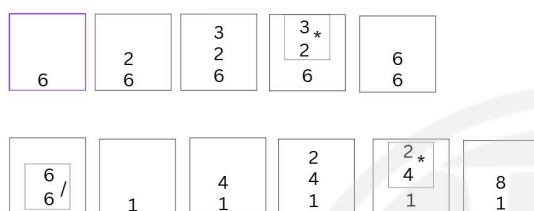
Step 6: Push 4 onto the stack → Stack [1, 4]

Step 7: Push 2 onto the stack → Stack [1, 4, 2]

Step 8: Encounter the * operator (Multiply the top elements, 4 and 2) → $4 * 2 = 8$

Push the result (8) back on the stack → Stack [1, 8]

After the second * is evaluated, the top elements of the stack are 8 and 1,



The Final Answer is: 8,1

Q25 Text Solution:

Explanation:

A binary search tree follows these properties:
For any node, all values in the left subtree are smaller than the node.

All values in the right subtree are greater than the node.

Step 1: Insert 60 → 60 becomes the root node.

Step 2: Insert 25 → Since 25 is less than 60, it goes to the left.

Step 3: Insert 72 → Since 72 is greater than 60, it goes to the right.

Step 4: Insert 15 → Since 15 is less than 60 and less than 25, it goes to the left of 25.

Step 5: Insert 30 → Since 30 is less than 60 but greater than 25, it goes to the right of 25.

Step 6: Insert 68 → Since 68 is less than 72 but greater than 60, it goes to the left of 72

Step 7: Insert 101 → Since 101 is greater than 72, it goes to the right of 72.

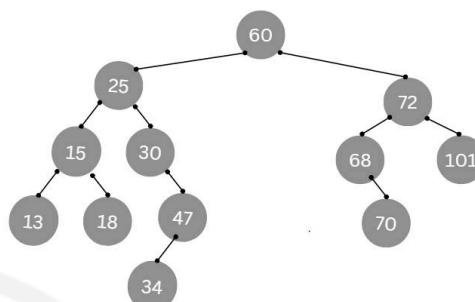
Step 8: Insert 13 → Since 13 is less than 60, less than 25, and less than 15, it goes to the left of 15.

Step 9: Insert 18 → Since 18 is less than 60, less than 25, and greater than 15, it goes to the right of 15.

Step 10: Insert 47 → Since 47 is less than 60, greater than 25, and greater than 30, it goes to the right of 30

Step 11: Insert 70 → Since 70 is less than 72, greater than 60, and greater than 68, it goes to the right of 68.

Step 12: Insert 34 → Since 34 is less than 60, greater than 25, greater than 30, but less than 47, goes to the left of 47.



The total number of nodes in the left subtree of the root (60) is: 25, 15, 13, 18, 30, 47, 34

The Final Answer is: 7

Q26 Text Solution:

Explanation:

Total Number of children:

Degree	Contribute child	Internal Node	Total Node
1	1	4	$1 \times 4 = 4$
2	2	3	$2 \times 3 = 6$
3	3	3	$3 \times 3 = 9$
	Total	= 10	= 19

Total Number of children: $4 + 6 + 9 = 19$

Total Number of Internal Node: $4 + 3 + 3 = 10$

Formula:

Number of leaf nodes = Total number of children – Total Number of Internal Node + 1

(where +1 accounts for the root node having one more child than its degree.)

Number of leaf nodes = $19 - 10 + 1$

Number of leaf nodes = 10

The Final Answer is: 10

Q27 Text Solution:

Correct Match:

(Graph Algorithm)	(Time Complexity)
(a) Dijkstra's algorithm	$O(V^2)$
(b) Kruskal's algorithm	$(E \log E)$



(c) Floyd-Warshall algorithm	$O(V^3)$
(d) Topological sorting	$(V+E)$

Q28 Text Solution:

The correct option is C.

A bipartite graph is a special kind of graph with the following properties-

- It consists of two sets of vertices X and Y.
- The vertices of set X join only with the vertices of set Y.
- The vertices within the same set do not join.

Bipartite graphs may be characterized in several different ways:

- A graph is bipartite if and only if it does not contain an odd cycle.
 - A graph is bipartite if and only if it is 2-colorable, (i.e. its chromatic number is less than or equal to 2).
 - The spectrum of a graph is symmetric if and only if it's a bipartite graph.
- Option (C) is false

Q29 Text Solution:

The correct option is B.

Let X is representing Which has to be delete Node

in a single linked list, We have only next pointer but not the previous pointer.

But we have find the previous node to X ==> which takes $O(n)$ in worst case, After finding it, Prev -> next = q-> next ;

free(q) ;

which takes $O(1)$ time only.

TC = $O(n) + O(1) = O(n)$.

Q30 Text Solution:

The correct option is C.

The second smallest element of n elements can be found with $n + \text{ceil}(\log n) - 2$ comparisons in the worst case.

If we have n elements, n-1 comparisons are required to find the minimum element.

During the procedure, each element is compared at most log n times. Because every time, we

compare the elements, at the second iteration list, the search will proceed in $n/2$ elements. Then in the next iteration, the search goes on for $n/4$ elements. In the end, there is only one element left. So, at most log n times each element is compared.

At the end, there are $n + \log n - 2$ comparisons to find the second minimum element.

Q31 Text Solution:

Option A is Correct

(i) $L(s) \subseteq L(r)$ and $L(s) \subseteq L(t)$:

$L(s) \subseteq L(r)$:

Every string in $L(s)$ starts with a, followed by zero or more a's, and ends with b. Strings in $L(r)$ can start with a, followed by any combination of a and b. Since $L(r)$ allows all strings in $L(s)$, $L(s) \subseteq L(r)$ is true.

$L(s) \subseteq L(t)$:

$L(s)$ includes strings like ab, aab, aaab, ... all of which fit the pattern of $t = a^*b$. Therefore, $L(s) \subseteq L(t)$ is true.

Thus, statement (i) is correct.

(ii) $L(r) \subseteq L(s)$ and $L(s) \subseteq L(t)$:

$L(r) \subseteq L(s)$:

$L(r)$ includes strings like ab, aab, abb, aabb, abab, ... $L(s)$ only allows strings starting with a, followed by a^* , and ending with b. For example, $abb \in L(r)$ but $abb \notin L(s)$. Therefore, $L(r) \subseteq L(s)$ is false.

$L(s) \subseteq L(t)$:

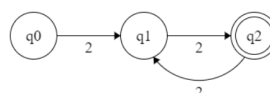
As shown earlier, $L(s) \subseteq L(t)$ is true.

Since one part of statement (ii) is false, the entire statement (ii) is false.

Q32 Text Solution:

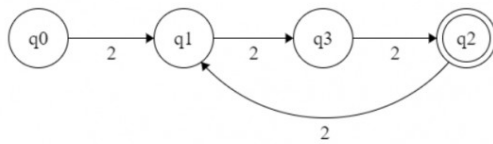
The correct option is B.

For $n = 2$, $\{2^{2k} \mid k > 0\} = \{2^2, 2^4, 2^6, \dots\}$, the automata for this language would be:



For $n = 3$, $\{2^{3k} \mid k > 0\} = \{2^3, 2^6, 2^9, \dots\}$, the automata for this language would be:





So, for $\{2^{nk} \mid k > 0\}$, the number of states should be $n+1$.

Q33 Text Solution:

The correct option is D.

String: "a d e f b g h n m p"

number of characters = $n = 10$

Repetition of the string is not present.

number of substring = $\frac{10*(10+1)}{2} + 1 = 56$

Q34 Text Solution:

The correct option is C.

If $y = 8$ and $|y| = 2|x|$, then $|x| = 4$ since it is finite, it will be accepted by a finite automaton.

If $y = 8$ and $|y| = |x|$ then $|x| = 8$ since it is finite, it will be accepted by a finite automaton.

Hence L_1 and L_2 both are accepted by finite automata. Therefore, they are regular language.

Q35 Text Solution:

The correct option is A.

Statement I: Correct

$L_1 = \{a^{n+m}b^n a^m \mid n, m \geq 0\}$

The above language is accepted by Deterministic Pushdown Automata. Hence it is a Context-free language.

Explanation:

- Push all a's that come as input and when b's come as input pop "n" number of a's for "n" b's from the top of the stack.
- When again a's come as input pop up all the remaining a's from the top of the stack for "m" number of a's coming as input. So we can have a Deterministic Pushdown Automata for L_1 so we can say that L_1 is a CFL.

Statement II - Incorrect

$L_2 = \{a^{n+m}b^n a^{n+m} \mid n, m \geq 0\}$

The above language is not accepted by

Deterministic Pushdown Automata. Hence it is not a Context-free language.

Q36 Text Solution:

The correct option is D.

- $\overline{L_1}$ is context-free.

CFLs are not closed under complementation.

So the first one is not correct.

- $L_1 - R$ is context-free.

It can also be written as $L_1 \cap \overline{R}$. Regular

languages are closed under

complementation. CFLs are closed under

intersections with regular language.

So $L_1 - R$ is context-free.

- $(\overline{L_1} \cap \overline{L_2}) - R$

CFLs are not closed under complementation and moreover not closed under intersection.

Q37 Text Solution:

The correct option is B.

Statement: I Decidable

A language is decidable when there is a machine that halts for every string in that language and goes to a final state for the string, not in the language. A finite automaton is a special case of a Turing machine which halts on all inputs that are in the language, so it is decidable.

Statement: II Undecidable

The regular property of context-free language is undecidable. This can be proved by taking regular language as Σ^* . Assume both G and R as part of the input problem. For any family of languages, whether it is equal to other languages is decidable if both L_1 and L_2 are subsets of each other.

Statement: III Undecidable

Turing machine is for recursive enumerable languages (for type 0 problems). Type 0 problems are undecidable. If we do not know about the Turing machine, it could be possible that the result for any two numbers will go into an infinite loop and never halt. If it happens, then the problem is undecidable.

Q38 Text Solution:



Option- **B**(Is $w \in L$, where w is a string ?)

Explanation: For recursive languages, the membership problem ($w \in L$) is decidable by definition. Recursive languages are ones where a Turing machine can determine if a given string (w) belongs to the language (L). As a result, this is a valid and solvable problem.

Q39 Text Solution:

Option- **D**(Neither S_1 nor S_2)

Explanation:

Step 1: Ambiguity Check: The grammar is ambiguous as the string c can be obtained from both A and B .

- $S \rightarrow A \rightarrow c$ - $S \rightarrow B \rightarrow c$.

This ambiguity causes parsing conflicts in both LL(1) and LR(1) parsers, as the parser cannot determine whether c belongs to A or B .

Step 2: LL(1) Parsing Analysis: An LL(1) parser predicts the next rule based on the current input symbol and the non-terminal being expanded, with only one lookahead token.

First and Following Sets:

1. First Sets

- $\text{First}(A) = \{a, c\}$
- $\text{First}(B) = \{b, c\}$

$\text{First}(S) = \text{First}(A) \cup \text{First}(B) = \{a, b, c\}$.

2. Following Sets: - $\text{Follow}(S) = \{\$ \}$ (assuming $\$$ is the end-of-input marker)

- $\text{Follow}(A) = \{\$ \}$
- $\text{Follow}(B) = \{\$ \}$

Conflict in LL(1) Table:

The terminal c appears in both $\text{First}(A)$ and $\text{First}(B)$.

When the input is c , the parser cannot decide whether to use $S \rightarrow A$ or $S \rightarrow B$.

Thus, the grammar is not LL(1) because it violates the LL(1) parsing condition (no conflicts in the parse table).

Step 3: LR(1) Parsing Analysis: An LR(1) parser is more powerful than LL(1). It works by maintaining a stack and using lookahead tokens to resolve conflicts. However, it cannot handle all ambiguous grammars.

Conflict in LR(1) Parsing:

The ambiguity in the grammar (e.g., c being

derivable from both A and B) leads to a shift-reduce conflict in the LR(1) parsing table:

When the parser sees c , it cannot decide whether to shift (read more input) or reduce (apply $A \rightarrow c$ or $B \rightarrow c$).

Thus, the grammar is also not LR(1) because ambiguity causes parsing conflicts.

Q40 Text Solution:

Options- **C**(Ambiguous)

Explanation: The given grammar is:

$$S \rightarrow (S) \mid SS \mid \epsilon$$

This grammar is **ambiguous** because certain strings can have multiple valid parse trees. For example:

1. Parse Tree 1:

$S \rightarrow SS$

First S

$\rightarrow (S) \rightarrow (\epsilon)$, *Second $S \rightarrow (S) \rightarrow (\epsilon)$*

2. Parse Tree 2:

$S \rightarrow (S)$, where S inside $(S) \rightarrow SS$, and each ϵ

Such ambiguity means the grammar cannot uniquely decide how to parse strings, making it unsuitable for predictive parsing, which requires unambiguous grammar.

Q41 Text Solution:

Option- **D**($2^{10} + 2^{11}$) bytes

Explanation:

- The frame buffer uses **10 bits per pixel**, which means it can represent $2^{10} = 1024$ different pixel values.
- Each color (R, G, B) requires **8 bits** (1 byte), so each color entry in the LUT takes **3 bytes** (24 bits).
- The LUT needs to store 1024 color values, so the total size of the LUT is:

$$1024 \times 3 = 3072 \text{ bytes}$$

- $3072 \text{ bytes} = 2^{10} + 2^{11} \text{ bytes}$

Q42 Text Solution:



Option- B

Explanation: The figure on the right is rotated anti-clockwise by 90 degrees. We know the Rotation matrix in the case of anti-clockwise

$$\text{rotation is: } A = \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix}$$

The option is given in a 3D matrix, so we have to transform the given 2D matrix into a 3D homogenous matrix system. So it is like,

$$\begin{bmatrix} \cos 90 & -\sin 90 & 0 \\ \sin 90 & \cos 90 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Here, we can see in the figure rotation is done concerning 2, so options are already given after rotating it w.r.t 2. We don't need to do it again.

So, the matrix after rotation w.r.t 2 looks like this:

$$\begin{bmatrix} 1 & -2 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Upto this by seeing the position of the - sign we can easily find that the correct option is either B or C. So, let's find it among these two now. We know that the Translation matrix is given by:

$$\begin{bmatrix} 1 & 1 & tx \\ 0 & 0 & ty \\ 0 & 0 & 1 \end{bmatrix}$$

See the figure the point which is at (0,0) is now transformed to 6 along the x-axis and 1 along y-axis. Therefore, tx = 6 & ty = 1, putting values we can get the matrix:

$$\begin{bmatrix} 1 & -2 & tx \\ 1 & 0 & ty \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -2 & 6 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Q43 Text Solution:

Option- B(P and Q Only)

Explanation: To evaluate the statements regarding the Bresenham line algorithm, let's analyze each statement one by one:

Statement P: For a line with slope $m > 1$, we should change the outer loop in line (4) to be over y .

True: This statement is correct. The Bresenham

algorithm works by iterating over the variable that changes more rapidly. If the slope $m > 1$ (steep line), we would iterate over y rather than x .

Statement Q: Lines (10) and (12) update the decision variable d through an incremental evaluation of the line equation f .

True: This statement is correct. The algorithm updates the decision variable incrementally based on the current position in the line-drawing process, effectively determining whether to increment y .

Q44 Text Solution:

Option- A(P and Q only)

Explanation:

P: Perspective projection simulates the way humans sense depth in the real world. Objects farther away from the observer (center of projection) appear smaller because the projection rays converge at the center.

This is a key feature of perspective projection.

Q: In parallel projection, the projection rays are parallel to one another, which means that the projection direction is independent of the object's position. This can be viewed as the spectator being infinitely far away so that the convergence of projection beams (as in perspective projection) is unimportant and the rays are actually parallel.

Q45 Text Solution:

Option- B(Q only)

Explanation: The Z-buffer approach stores the distance from the camera by assigning a depth value (z-coordinate) to each pixel in the frame buffer. This value is often expressed in 16-bit, 32-bit, or 64-bit precision, depending on the implementation.

During rendering, the Z-buffer compares the depth values of surfaces at each pixel to determine which one is visible. Thus, Q is correct.

Q46 Text Solution:

Option- B($p=2^{k+1}-1$ and $0 \leq k < m$)

Explanation:

Step-by-Step Explanation:

p accumulates the total of powers of two: $p=2^0$



$+2^1 + 2^2 + \dots + 2^{k-1}$. In each repetition, k grows by 1.

Mathematical formula for "p": The total of powers of 2 up to 2^{k-1} is $p = 2^k - 1$.

This describes p at the beginning of the k -th iteration.

B. Before each iteration begins, p can also be expressed as $p = 2^{k+1} - 1$, which predicts the final sum when the current iteration is done. The loop is executed while $k < m$, assuring $0 \leq k < m$.

Verification using an Example: In the initial state ($k=0$), $p=0$, and $2^{0+1}-1=1-1=0$ (correct)

First iteration ($k=1$): $p=2^1=2$, and $2^{1+1}-1=4-1=3$, which is correct after the current step.

Thus, **Option B is correct** because it describes the state of p at the end of each iteration while ensuring the condition $0 \leq k < m$.

Q47 Text Solution:

Option- **A(Only S1 is true)**

Explanation: Explanation of Program P1:

1. Before the function call: $i = 0$.

2. Inside the function f:

Changing a to 1 does not affect i because it is given as a value. $b = 2$ converts i to 2 since b points to i .

$c = 3$ alters i by reference.

3. After the function call: The output is 3.

4. Output: The program outputs i , which is three.

Thus, $S1$ is correct: $P1$ prints 3.

Explanation of Program P2

In this, $f(a)$ calls the function f which is:

```
double &f(double &d)
{
    d=4; // it will update the value of a to 4 and
    returns a reference to double b.
    return b; // here it returns the value of b as 5.
}
So, in main
int main ()
{
    f(a)=5; // a becomes 4
    cout <<a<< " "<b; // it prints 4: 5
}
```

Q48 Text Solution:

Explanation:

Recursive call: $f(m*2, n*3)$

Result: $\text{result} = (\text{float } m) / (\text{float } n)$;

Call	m	n	float(m) / float (n)	Result	Recursive call
First call: f(2,3)	2	3	2.0 / 3.0	0.6667	f(4,9)
Second call: f(4,9)	4	9	4.0 / 9.0	0.4444	f(8,27)
Third call: f(8,27)	8	27	8.0 / 27.0	0.2963	f(16,81)
Fourth call: f(16,81)	16	81	16.0/81.0	0.1975	f(32,243)
Fifth call: f(32,243)	32	243	32.0/243.0	0.1317	f(64,729)
Sixth call: f(64,729)	64	729	64.0/729.0	0.0878	f(128,2187)
Seventh call: f(128,2187)	128	2187	128.0/2187.0	0.0586	f(256,6561)
Eighth call: f(256,6561)	256	6561	256.0/6561.0	0.0390	f(512,19683)
Ninth call: f(512,19683)	512	19683	512.0/19683.0	0.0260	f(1024,59049)
Tenth call: f(1024,59049)	1024	59049	1024.0/59049.0	0.0173	...

Result : $0.6667 + 0.4444 + 0.2963 + 0.1975 + 0.1317 + 0.0878 + 0.0586 + 0.0390 + 0.0260 = 1.948 \approx 2$

The Final Answer is: 2

Q49 Text Solution:

Explanation:

Initially value of $b=0$

Consider $a=5$

Iteration 1:

for(int $i=0; i<32; i++$) // condition is true

$b=b<<1$; // b becomes $0(0 \times 2 = 0)$

$b=b|(a\&1)$; // $0|(5\&1)$; b becomes 1

$a=a>>1$; // as this is a logical shift; so a becomes 2

Iteration 2 :

for(int $i=0; i<32; i++$) // i is 1, condition is true

$b=b<<1$; // b becomes 2

$b=b|(a\&1)$; // $2|(2\&1)$; b becomes 2

$a = a >> 1$; // as this is a logical shift; so a



becomes 1

Iteration 3:

for(int i=0; i<32; i++) // i is 2, condition is true

b=b<<1; // b becomes 4

b=b|(a&P); // 4|(1 & 1); b becomes 5

a = a >>> 1; // as this is a logical shift; so a

becomes 0

Iteration 4:

for(int i=0; i<32; i++) // i is 3, condition is true

b = b<<1; // b becomes 10

b= b|(a & 1); // 10|(0 & 1); b becomes 10

a = a >>> 1; // as this is a logical shift; so a

becomes 0

After this, value of b will multiply by 2 on every iteration

when i=4, " b=20

when i=5, " b=40

when i=6, b=80=2³×10

when i=7, " b=160= 2⁴ × 10

.....

when i=30, b=2²⁷×10

but Size of int is 4 byte in java: Range -2³¹ to 2³¹-1

when i=31, b=2²⁸×10=-1610612736

a=5=(00000000...0101)₂

reverse of

a=101000000000...0000)₂=-1610612736

Therefore, return the int that has the reversed binary representation of integer a.

The Final Answer is: Option B

Q50 Text Solution:

Explanation:

<td style="text-align : center; vertical-align : middle;"> BASH </td>

Horizontal alignment: text-align: center; is the correct CSS property for horizontally centering text within a <td>

Vertical alignment: vertical-align: middle; is valid and aligns the text vertically in the middle of the cell.

The Final Answer is: Option D

Q51 Text Solution:

Explanation::

int x = strlen(w); →Get the length of the string

Loop through half of the string to swap characters.

for (int i = 0; i < x ; i++) {

c = w[i]; → Store the current character

w[i] = w[x - i - 1]; → Swap the character with its

corresponding pair from the other end

w[x - i - 1] = c; →Complete the swap

It outputs the contents of the array in the original order.

The Final Answer is: Option B

Q52 Text Solution:

Explanation:

The Software Requirement Specification (SRS) is considered consistent.

This happens when all individual requirements described in it do not conflict with each other.

If the requirements meet these conditions, the SRS is considered consistent.

The Final Answer is: Option B

Q53 Text Solution:

Explanation:

Perfective maintenance involves making changes to improve software performance or add new features based on user feedback.

It focuses on enhancing functionality to meet evolving user needs.

The Final Answer is: Option C

Q54 Text Solution:

Explanation:

(a) State Diagram → (iv) Used to describe the dynamic behaviour of objects and could also be used to describe the entire system's behaviour.

(b) Use-Case Diagram → (i) Describes how the external entities (people, devices) can interact with the system.

(c) Class Diagram → (ii) Used to describe the static or structural view of a system.

(d) Activity Diagram → (iii) Used to show the flow of a business process, the steps of a use-case or the logic of an object's behaviour.

The Final Answer is: Option D

Q55 Text Solution:

Explanation:

Clean room software engineering model is an incremental software paradigm for the



development of zero defect and high reliability software using box structure specification, statistical quality control and certification. It is a practical development and certification approach that uses statistical quality control to reduce software defects, to overcome unconscious copying of copyrighted code.

The Final Answer is: Option D

Q56 Text Solution:

Option A is Correct

- Statement Coverage (P): 3 test cases are sufficient to execute all statements in the program at least once. This includes:
 - $m < 0$ branch to execute the statement $res = n - m$,
 - $m \geq 0, x || y = \text{True}$, and $n \neq m$,
 - $m \geq 0, x || y = \text{False}$, to cover all remaining statements.
- Branch Coverage (Q): 4 test cases are required to exercise all decision outcomes (both true and false) at the branches:
 - $m < 0$: True and False.
 - $x || y$: True and False.
 - $n == m$: True and False.

Thus, the value of (P,Q) is (3, 4).

Q57 Text Solution:

Explanation:

Total number of modules in the system = 940

Changed modules = 90

New modules added = 40

Old modules removed = 12

Unmodified modules = Total modules - (Changed modules + New modules added + Removed modules)

$$\Rightarrow 940 - (90 + 40 + 12) = 940 -$$

$$142 = 798.$$

$$\text{Software Maturity Index} =$$

$$= \frac{\text{Number of modules unmodified}}{\text{Total number of modules in the system}} \\ \frac{798}{940} \approx 0.849$$

The Final Answer is: Option A

Q58 Text Solution:

Explanation:

Software Reengineering involves improving existing software by addressing issues like poor design, failure rates, and code structure, without changing the software's external behavior.

Reverse Engineering is the process of analyzing software to understand its design and extract specifications, typically for updating, improving, or documenting legacy systems.

The Final Answer is: Option C

Q59 Text Solution:

Explanation:

Following the plan: Agile de-emphasizes rigidly following a set plan. Instead, it focuses on flexibility and adapting to change as the project progresses.

The Final Answer is: Option D

Q60 Text Solution:

Explanation:

Common coupling occurs when two or more modules have access to shared global data. Changes to this data can affect multiple modules, leading to tight interdependencies.

Content coupling is the highest (tightest) form of coupling, where one module directly accesses or modifies the internal details (like variables or logic) of another module.

The Final Answer is: Option B

Q61 Text Solution:

Explanation:

	List I		List II
(a)	Blowfish	(iii)	1- 448 bits
(b)	DES	(iv)	56 bits
(c)	IDEA	(ii)	128 bits
(d)	RC5	(i)	128 - 256 bits

The Final Answer is: (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i).

Q62 Text Solution:

An IP address is a 32-bit address, typically represented as four bytes (or 4 octets). It is divided into two parts:

- Network Address: This part identifies the network on which a device is located. The bits used for the network address depend on the class of the IP address (Class A, B, C, etc.).



2. **Host Address:** This part identifies the specific device or host within that network.

For example, in a Class C IP address (such as 192.168.1.1), the first three octets (e.g., 192.168.1) represent the Network Address, while the last octet (1) represents the Host Address.

Q63 Text Solution:

In the **ISO-OSI reference model**, the **Session Layer** (Layer 5) is responsible for managing and controlling the dialog between two devices.

1. **Dialog Control:** Manages the conversation between the two devices, determining if communication is half-duplex (one-way) or full-duplex (both-way).
2. **Token Management:** Ensures that only one device can send data at a time, avoiding conflicts (used in protocols like Token Ring).
3. **Synchronization:** Provides checkpoints and recovery mechanisms during communication. It ensures that data can be recovered if there is an interruption, and it maintains the order and integrity of the communication stream.

Option A is correct.

Q64 Text Solution:

SaaS (Software as a Service) is a cloud computing model in which the service provider delivers software applications over the internet. These applications are hosted on the provider's infrastructure and are made available to users on a subscription basis.

Examples of SaaS include email services like Gmail, office software like Google Docs or Microsoft Office 365, and CRM tools like Salesforce.

SaaS provides complete software solutions where the user only needs to interact with the application. The provider manages everything else, including operating systems and application software.

Q65 Text Solution:

Broadband CDMA is the technology used in **Third Generation (3G)** mobile networks. It refers to **CDMA2000**, which is a broadband version of CDMA used to provide higher data rates suitable

for 3G networks. Broadband CDMA supports voice, data, and multimedia services at significantly higher speeds compared to earlier 2G technologies.
Option D is correct.

Q66 Text Solution:

i) Firewalls can screen traffic going into or out of an organization.

True: A **firewall** is a network security device that monitors and controls incoming and outgoing network traffic based on predetermined security rules. Firewalls can filter traffic in both directions: traffic coming into the network and traffic going out of the network.

(ii) Virtual private networks can simulate an old leased network to provide certain desirable properties.

True: A **Virtual Private Network (VPN)** is a technology that allows private networks to be extended over public networks (such as the internet). A VPN can simulate a leased line by creating a secure tunnel between two endpoints, providing privacy, security, and reliable communication similar to the old, dedicated leased lines.

Q67 Text Solution:

(i) Windows XP supports both peer-to-peer and client-server networks.

True. Windows XP is capable of supporting both **peer-to-peer (P2P)** and **client-server** networking configurations. In a **peer-to-peer network**, computers act as both clients and servers, sharing resources without a central server. In a **client-server network**, a dedicated server manages resources and services, and the other devices act as clients. Windows XP could be configured to work in either mode, depending on the network setup.

(ii) Windows XP implements Transport protocols as drivers that can be loaded and unloaded from the system dynamically.

True. In Windows XP, **transport protocols** (like TCP/IP) are implemented as **protocol drivers**. These drivers are responsible for communication over the network. They can be dynamically loaded and unloaded, allowing the system to add



or remove networking protocols as needed without rebooting the system.

Q68 Text Solution:

In Linux, the `lpr` command is used to print files. It sends files to the default printer for printing. The `lpr` command is part of the **Line Printer Daemon (LPD)** system, which manages printing in Unix-like systems.

Option D is correct.

Q69 Text Solution:

In the Readers-Writers problem, the goal is to manage access to shared data in a way that allows multiple readers to access the data simultaneously, but ensures that writers have exclusive access to the data (i.e., no readers or other writers can access the data when a writer is working on it).

i) Writers are given exclusive access to shared objects.

True: In the Readers-Writers problem, writers must have exclusive access to the shared resource because writing modifies the data, and concurrent writes or reads during a write can lead to inconsistent or corrupted data.

Option A is correct.

Q70 Text Solution:

Available Resources: $R=12$

Max needs per process: $m=4$

Deadlock can occur if processes are unable to acquire the needed resources to proceed. To prevent this:

At least one process must always be able to complete and release its resources.

The **minimum resources needed to avoid deadlock** are calculated as: $n \times (m-1) + 1 \leq R$ where:

n = number of processes,

m = maximum resources needed per process,

R = total available resources.

Calculation:

Substitute the given values ($R=12, m=4$):

$$n \times (4-1) + 1 \leq 12$$

$$3n + 1 \leq 12$$

Solve for n :

$$3n \leq 11$$

$$n \leq 11/3 = 3.67$$

Since n must be an integer, the maximum number of processes is:

$$n_{\max} = 3$$

Option A is correct.

Q71 Text Solution:

The **dirty bit** is a flag in the page table entry of a memory management system.

It is used to indicate whether a page in memory has been **modified** (written to) after it was loaded from disk into memory.

If the dirty bit is set (value = 1), it means the page has been modified and must be written back to the disk before being replaced or removed from memory.

If the dirty bit is not set (value = 0), it means the page has not been modified, and there is no need to write it back to disk when it is removed.

Purpose of the Dirty Bit:

It improves **efficiency** in memory management because it avoids unnecessary writes to disk for pages that haven't been modified.

It is commonly used in **virtual memory systems** and caching mechanisms.

Option D is correct.

Q72 Text Solution:

Initially,

The requests are :

1 - 1 | 2

2 - 1 | 2

4 - 1 | 2 | 4, Pagefault = 1

5 - 2 | 4 | 5, Pagefault = 2

2 - 2 | 4 | 5, Pagefault = 2

1 - 4 | 5 | 1, Pagefault = 3

2 - 5 | 1 | 2, Pagefault = 4

4 - 1 | 2 | 4, Pagefault = 5

Option C is correct.

Q73 Text Solution:

1. First Round of Execution:

Each process gets **4 ms** (time quantum) or its burst time (if less than 4 ms):

- P_1 : $5 - 4 = 1$ ms remaining
- P_2 : $7 - 4 = 3$ ms remaining
- P_3 : $6 - 4 = 2$ ms remaining
- P_4 : $4 - 4 = 0$ ms (completed)



At the end of the first round, P_4 finishes its execution.

2. Waiting Time for P_4 :

- P_4 starts execution immediately after P_3 completes its first quantum.
- Time spent waiting:
- P_1 : 4 ms
 - P_2 : 4 ms
 - P_3 : 4 ms
- Total waiting time for P_4 : $WT_{P_4} = 4+4+4 = 12\text{ms}$.

Q74 Text Solution:

The correct option is B.

A mutual exclusion (mutex) is a program object that prevents simultaneous access to a shared resource.

A critical section is a code segment that accesses shared variables and has to be executed as an atomic action. The critical section problem refers to the problem of how to ensure that at most one process is executing its critical section at a given time.

ENTRY SECTION
Process Q CRITICAL SECTION
EXIT SECTION
REMAINING SECTION

Since process 'Q' is in the critical section then other processes P and R cannot be in the critical section. Therefore neither 'P' nor 'R' executes in their critical section.

Q75 Text Solution:

The correct option is C.

Data structure used in Queue, implemented using linked list where processes are kept.

Type of Queue	Description
---------------	-------------

Job queue	When the process enters the system (secondary Memory), it is kept in the job queue.
Ready queue	Process waiting in the system (Main Memory) ready to execute is kept in the Ready queue.
Device queue	Anything related to input or output

Q76 Text Solution:

The correct option is C.

- In Unix OS, the fork system call is used to create a new process where a process is a copy of itself, which is called the child process.
- The fork system call is usually a system call, implemented in the kernel.
- After a new child process is created, both processes will execute the next instruction following the fork() system call.

Q77 Text Solution:

The correct option is C.

Char and varchar types are similar but differ in their storage and retrieval ways. They also have different maximum lengths.

When char is declared with a length then its length is fixed with that is declared during the creation of the table. Here, char(20) is given for the value "Imnop" Length is fixed in char. So, attribute B has 20 spaces.

Varchar: Values in varchar columns are of variable length. Here, varchar(20) has the value "XYZ" for attribute A. So, attribute A takes 3 spaces.

Q78 Text Solution:

The correct option is A.

A warehouse generates summary data, which can be lightly or highly summarized. This data is updated as new data loads into the warehouse. The main purpose of summary data is to speed up query performance.

Q79 Text Solution:

The correct option is A.



The drop table deletes the whole structure of the relation and removes the table which cannot be obtained again.

Q80 Text Solution:

The correct option is C.

The first statement is correct. Hadoop clusters help in organizing and analyzing the data in a computational environment. It boosts the speed of data analysis. These clusters help in increasing the throughput performance. They are also resistant to any failure and data loss as data backup is maintained on the clusters to support redundancy. As per a research study in 2013, Facebook has become the world's largest Hadoop cluster storing more than hundreds of millions of gigabytes.

The second statement is also correct. Hadoop 2.0 has many advantages over the previous version. These are:

- Hadoop 2.0 allows live stream processing of real-time data.
- Its ability to process Terabytes and petabytes of data available in HDFS using MPI.
- It enables multi-tenancy support in Hadoop.

Q81 Text Solution:

The correct option is D.

Data scrubbing also known as data cleansing deals with detecting and removing errors and inconsistencies from data in order to improve the quality of data. Data quality problems are present in single data collection such as files and databases. It is a process to upgrade the quality of data before it is moved into a database warehouse.

Q82 Text Solution:

Option D is Correct

A candidate key is the minimal set of attributes that can uniquely determine all attributes in R.

Step 1: Attribute Closure Method

To find candidate keys, we compute the closure of attribute sets (starting from smaller sets) and check if the closure includes all attributes in R.

Closure of Single Attributes:

1. Closure of A:

- From $A \rightarrow B$: Add B.
- From $D \rightarrow A$: No new attributes added.
- Closure of $A = \{A, B\}$. (Does not include all attributes in R.)

2. Closure of B:

- From $B, C \rightarrow D$: B alone cannot determine anything.
- Closure of $B = \{B\}$. (Does not include all attributes in R.)

3. Closure of C:

- From $E \rightarrow C$: C alone cannot determine anything.
- Closure of $C = \{C\}$. (Does not include all attributes in R.)

4. Closure of E:

- From $E \rightarrow C$: Add C
- Closure of $E = \{E, C\}$. (Does not include F.)

5. Closure of F:

- No FDs depend on F.
- Closure of $F = \{F\}$. (Does not include all attributes in R.)

Combining Attributes:

We now try combinations of attributes to cover all attributes $\{A, B, C, D, E, F\}$.

Testing AE:

- Closure of AE:
 - From $E \rightarrow C$: Add C.
 - From $B, C \rightarrow D$: Add D.
 - From $D \rightarrow A$: A is already present.
 - From $A \rightarrow B$: Add B.
 - Closure of $AE = \{A, B, C, D, E\}$.
 - Does not include F.

Testing BE:

- Closure of BE:
 - From $E \rightarrow C$: Add C.
 - From $B, C \rightarrow D$: Add D.
 - From $D \rightarrow A$: Add A.
 - From $A \rightarrow B$: B is already present.
 - Closure of $BE = \{A, B, C, D, E\}$.
 - Does not include F.



Testing AEF:

- Closure of AEF:
 - From $E \rightarrow C$: Add C.
 - From $B, C \rightarrow D$: Add D.
 - From $D \rightarrow A$: A is already present.
 - From $A \rightarrow B$: Add B.
 - Closure of $AEF = \{A, B, C, D, E, F\}$.
 - Includes all attributes. AEF is a candidate key.

Testing BEF:

- Closure of BEF:
 - From $E \rightarrow C$: Add C.
 - From $B, C \rightarrow D$: Add D.
 - From $D \rightarrow A$: Add A.
 - From $A \rightarrow B$: B is already present.
 - Closure of $BEF = \{A, B, C, D, E, F\}$.
 - Includes all attributes. BEF is a candidate key.

Testing DEF:

- Closure of DEF:
 - From $D \rightarrow A$: Add A.
 - From $E \rightarrow C$: Add C.
 - From $A \rightarrow B$: Add B.
 - Closure of $DEF = \{A, B, C, D, E, F\}$.
 - Includes all attributes. DEF is a candidate key.

The candidate keys are: AEF, BEF, DEF.

Q83 Text Solution:

The correct option is B.

To be able to recover from failures that affect transactions, the system maintains a log to keep track of all transactions so as to recover the data. In a log-based recovery mechanism, redo operations are performed for those transactions that contain both start and commit log records. Undo operation is performed for the transactions in which a system crash happens before their commit operation.

As, here system crashes out just before log record (vii) is written. Before this, log record (vi) has not been committed. So, there is a need to undo record (vi) and redo log records (ii) and (iii)

Q84 Text Solution:

Step-by-Step Execution:

1. Join Operation:

The query joins the Students and Performance tables on the Roll-No column.

This results in the following intermediate table:

Roll-No	Name	Course	Marks
18CS101	Ramesh	DBMS	60
18CS101	Ramesh	Compiler Design	65
18CS102	Mukesh	DBMS	80
18CS102	Mukesh	Compiler Design	75
18CS103	Ramesh	DBMS	85
18CS103	Ramesh	Operating System	70

2. Grouping and Aggregation:

S.Name groups the results and the marks are summed for each group.

Grouping by S.Name results in two groups: "Ramesh" and "Mukesh".

3. Aggregation:

For Ramesh: Marks: $60 + 65 + 85 + 70 = 280$

For Mukesh: Marks: $80 + 75 = 155$

4. Final Result:

The query returns:

Name	SUM(Marks)
Ramesh	280
Mukesh	155

The query returns **2 rows**: one for "Ramesh" and one for "Mukesh".

Q85 Text Solution:

- Non-key: The field is not a primary key, meaning it can have duplicate values.
- Ordering: The field is used to order the records in the table.

In a clustering index, the data is physically ordered based on the non-key field, which helps in efficiently retrieving records that have the same value for the clustering field.

Q86 Text Solution:

Ethernet address is basically the MAC address, which is supposed to be unique to a NIC. Thus it



is unique over the Internet.

Q87 Text Solution:

1. Total Addresses:

- /20 provides ($2^{12} = 4096$) addresses.

2. Organization A:

- Needs half: ($4096 / 2 = 2048$) addresses.
- Subnet mask: /21 (2048 addresses).
- Valid subnet: 245.248.136.0/21.

3. Organization B:

- Needs a quarter: ($4096 / 4 = 1024$) addresses.
- Subnet mask: /22 (1024 addresses).
- Valid subnet: 245.248.128.0/22.

Conclusion:

- **Organization A:** 245.248.136.0/21
- **Organization B:** 245.248.128.0/22

This allocation ensures no overlap and fits within the original block.

Q88 Text Solution:

- **S1:** TCP uses the TCP window for flow control and the congestion window for congestion control.
- **S2:** UDP does not have fields in its header to control flow or congestion.

Therefore, the correct option is:

S1 is correct but S2 is not correct

Q89 Text Solution:

DNS Query: The client resolves the domain name to an IP address.

TCP SYN: The client initiates a TCP connection with the server.

HTTP GET Request: The client requests the webpage.

Correct Sequence:

DNS query, TCP SYN, HTTP GET request

This sequence ensures the domain is resolved, a connection is established, and then the webpage is requested.

Q90 Text Solution:

In symmetric key cryptography, both parties share the key. This means each person in a group

of (N) people needs a unique key to communicate with (N-1) others. The total number of keys required is similar to the number of edges in a complete graph with (N) vertices, which is $N(N-1)/2$.

Q91 Text Solution:

An agent can improve by saving the previous states on which it was earlier, hence in future it can learn to respond in the same situation better.

Q92 Text Solution:

In a semi-dynamic environment, the environment itself remains unchanged over time, but the agent's performance score can change. This means that while the agent is making decisions or taking actions, the surroundings stay the same, but the effectiveness or success of the agent's actions can vary.

For example, imagine a robot in a warehouse where the layout of the warehouse doesn't change, but the robot's efficiency in picking items might decrease over time due to battery depletion or wear and tear. The environment (warehouse layout) is static, but the performance score (efficiency) changes.

Therefore, the correct statement is:

- The environment itself does not change with the passage of time but the agent's performance score does.

Q93 Text Solution:

Explanation:

BFS search $\rightarrow O(b^d)$

DFS search $\rightarrow O(bm)$

Depth-limited search $\rightarrow O(b^l)$

Iterative deepening search $\rightarrow O(bd)$

The Final Answer is: (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i).

Q94 Text Solution:

Explanation:

	List I		List II
(a)	Greedy Best-First Search	(i)	Select a node for expansion if the optimal path to that node has been found.



(b)	A* Search	(iv)	Time complexity depends on the quality of the heuristic.
(c)	Recursive Best-First Search	(iii)	Suffers from excessive node generation.
(d)	Iterative-deepening A* Search	(ii)	Avoids substantial overhead associated with keeping the sorted queue of nodes.

Q95 Text Solution:

Explanation:

S1: An admissible heuristic never overestimates the true cost to the goal, ensuring $h(n) \leq h^*(n)$.

S2: A Monotonic(consistent) heuristic satisfies the triangle inequality: $h(n) \leq c(n, n') + h(n')$

where n is the current node being evaluated in the search process and n' is the neighboring node directly connected to n in the search graph.

The Final Answer is that both statements S1 and S2 are true.

Q96 Text Solution:

Explanation:

A solution in AND-OR search is a subtree with goal nodes at every leaf, solving the problem under all conditions. $\rightarrow$ True.

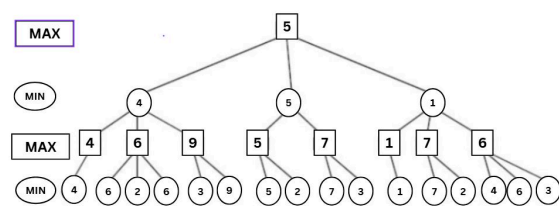
OR nodes represent choices, similar to branching in deterministic environments where one action has a specific outcome. $\rightarrow$ True.

AND nodes handle all possible outcomes of an action, like branching in non-deterministic environments. $\rightarrow$ True.

The Final Answer is: S1 - True, S2 - True, S3 - True

Q97 Text Solution:

Explanation:



The Final Answer is: 5

Q98 Text Solution:

Explanation:

Step 1: $\neg A \vee \neg B \vee \neg C \vee \neg D$ is a disjunction where at least one negated proposition must be true for the sentence to be true.

Step 2: with 4 prepositions (A, B, C, D) where 2^4 possible equals 16 Possible truth assignments.

Step 3: The sentence is true if at least one of the original propositions (A, B, C, or D) is false (because its negation would then be true)

Step 4: The sentence is false only when all propositions (A, B, C, D) are true, making the negations false.

i.e. as $A \vee B \vee C \vee D = (T \vee T \vee T \vee T \rightarrow T)$ and $\neg A \vee \neg B \vee \neg C \vee \neg D = (F \vee F \vee F \vee F \rightarrow F)$

Out of the 16 models, only 1 model makes the sentence false, so 15 models make it true.

The Final Answer is: 15

Q99 Text Solution:

Explanation:

There is a country that borders both India and Nepal.

Existential quantifier : $(\exists)$

Predicate : Country(c)

The relation "borders": The sentence mentions "borders both India and Nepal," implying that the country borders both India and Nepal.

Conjunction: The country borders both India and Nepal, which implies a logical conjunction (AND) between the two border relations.

It states that there exists a country c such that c is a country and borders both India and Nepal.

The Final Answer is: Option A

Q100 Text Solution:

Explanation:

Formula: **Conditional Probability of A given B:**

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

$$P(\text{Cavity} | \text{Toothache})$$

$$= \frac{P(\text{Cavity} \cap \text{Toothache})}{P(\text{Toothache})} = \frac{0.12}{0.2} = 0.6$$

$$P(\text{Cavity} \cap \text{Toothache}) = 0.108 + 0.012 = 0.12$$

$$P(\neg \text{Cavity} \cap \text{Toothache}) = 0.016 + 0.064 = 0.08$$

$$P(\text{Toothache}) = 0.108 + 0.012 + 0.016 + 0.064 = 0.2$$



$$P(\neg Cavity \mid Toothache) \\ = \frac{P(\neg Cavity \cap Toothache)}{P(Toothache)} = \frac{0.08}{0.2} = 0.4$$

Conditional Probability of Cavity Given

Toothache: 0.6

Conditional Probability of $\neg$ Cavity Given

Toothache: 0.4

The Final Answer is: <0.6,0.4>



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